FACULTY OF COMPUTING
SEMESTER 2 2023/2024

SECR1033 COMPUTER ORGANIZATION & ARCHITECTURE

SECTION 01

CPU & Hardware Performance Benchmarks

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Video Link:
https://drive.google.com/drive/folders/1POSki3l-PZJuTFn36cOJwQBcOcM649Yb?usp=sharing

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1.0 Introduction

1.1 Problem background

In today's rapidly evolving technological landscape, the performance of computer systems is paramount for both personal and professional applications. One critical component that significantly impacts overall system performance is the Central Processing Unit (CPU). Understanding and evaluating the performance of a CPU is essential for optimizing tasks, enhancing user experiences, and ensuring that systems meet the demands of various applications.

CPU benchmarking is a methodical process that involves running standardized tests to assess the performance capabilities of a CPU. These tests simulate different workloads and measure various aspects of CPU performance, such as processing speed, multitasking efficiency, and thermal management. By performing these benchmarks, users can gain valuable insights into the strengths and weaknesses of their CPUs, compare performance across different systems, and make informed decisions about hardware upgrades or optimizations. By using standardized tests and tools, we can measure and compare the efficiency, speed, and overall performance of CPUs.

In this project, we are focusing on using benchmarking tools such as CPU-Z, Geekbench, MiniTools, and Task Manager to assess and analyze the performance of various CPUs. Additionally, we are also going to develop an assembly language program to benchmark software using a specific polynomial equation, providing practical experience in low-level programming and performance measurement.

1.2 Aim

The aim of this project is to analyze and compare the performance of different computer systems using CPU-Z, Geekbench, MiniTools, and Task Manager. By collecting detailed hardware specifications and execution results, the project seeks to evaluate CPU performance, measure execution efficiency under varying workloads, and provide data-driven insights to inform hardware upgrade and optimization decisions. Ultimately, the project aims to enhance understanding of how various CPU attributes impact system performance and efficiency, contributing to better-informed choices in computer architecture and optimization strategies.

1.3 Objectives

- To understand the principles of CPU benchmarking
- To compare the performance of difference of CPUs
- To analyze the impact of different workloads on CPU performance
- To learn on how to use the free benchmarking tools and interpret their results

1.4 Scopes (Computer specification & benchmarking tools features)

Specification	Computer Type		
	VivoBook_ASUSLaptop	LENOVO 82K1	Dell Inspiron 16
Processor Type	Intel Core i5 1135G7	Intel Core i5 11300H	Intel Core i5 11400H
RAM Size	16 GBytes	16 GBytes	16 GBytes
GPU Type	NVIDIA GeForce MX330	NVIDIA GeForce GTX 1650	NVIDIA GeForce RTX 3050 Laptop GPU
Motherboard	ASUSTeK COMPUTER INC. X515EP	LENOVO LNVNB161216	Dell Inc. 1VFRT8
Bus Specs.	PCI-Express 4.0 (16.0GT/s)	PCI-Express 3.0 (8.0 GT/s)	PCI-Express 4.0 (16.0 GT/s)

2.0 Part 1
CPU Benchmarking and Performance Analysis Using Free Tools

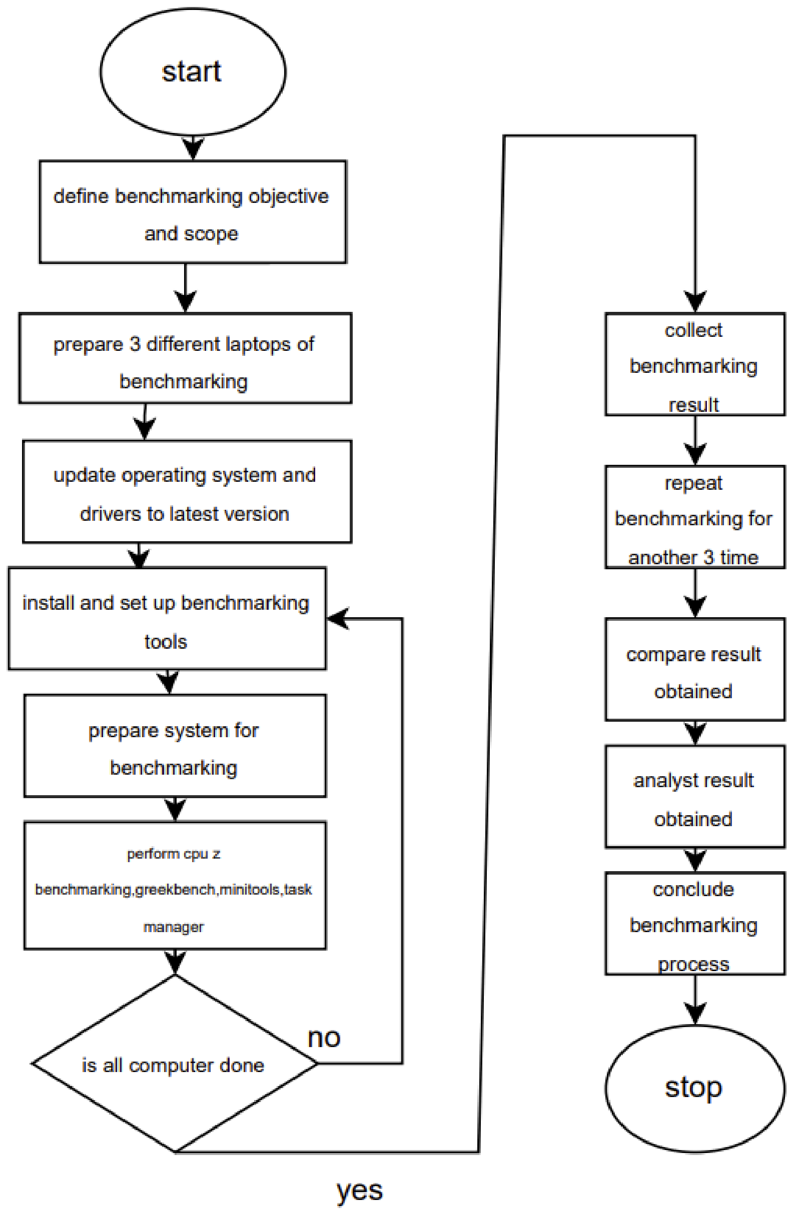
2.1 Overview

This project's initial phase explores CPU benchmarking to compare laptop performance. We'll use popular tools like Cinebench and CPU-Z to evaluate CPUs.

Cinebench runs standardized tests to provide comprehensive performance metrics. CPU-Z monitors key parameters such as clock speed, cache size, and core count. These tools enable objective assessment of CPU capabilities, facilitating informed decisions about system upgrades and optimizations.Our goals for this phase include,deepening our understanding of CPU benchmarking methods,gaining insights into performance evaluation using Cinebench and CPU-Z,developing proficiency in benchmarking software usage.By examining these tools, we'll enhance our ability to measure and analyze CPU performance effectively.

2.2 Flowchart

Figure 2.2: procedure of conducting benchmarking

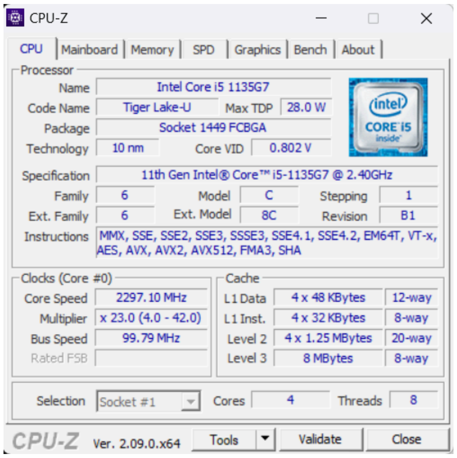


2.3 Benchmarking Results

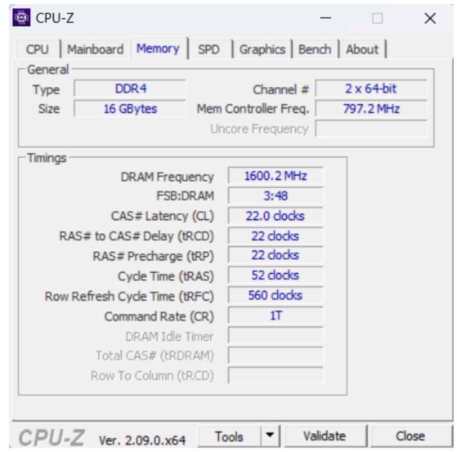
Computer A (CHIN PEI WEN)

CPUZ (System Info)

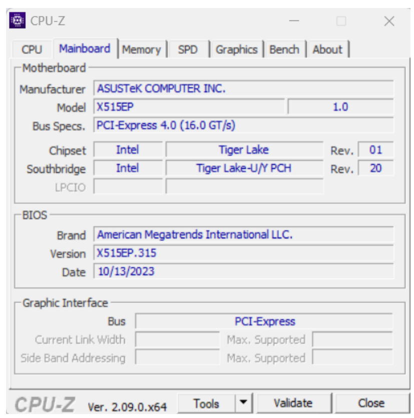
CPU Processor and cache :



Memory :

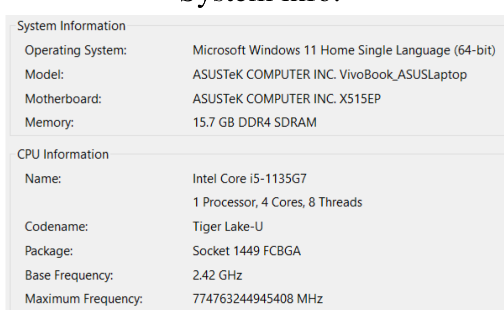


Motherboard :

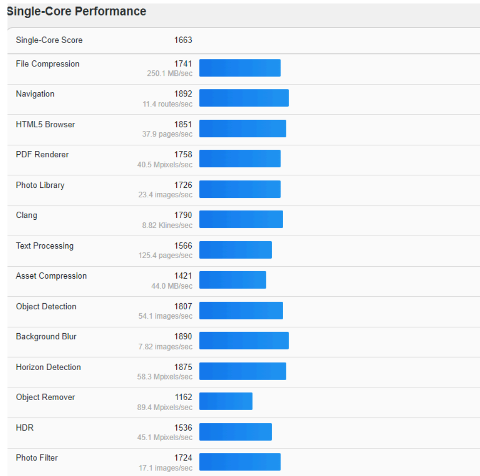


Geekbench (CPU stress test)

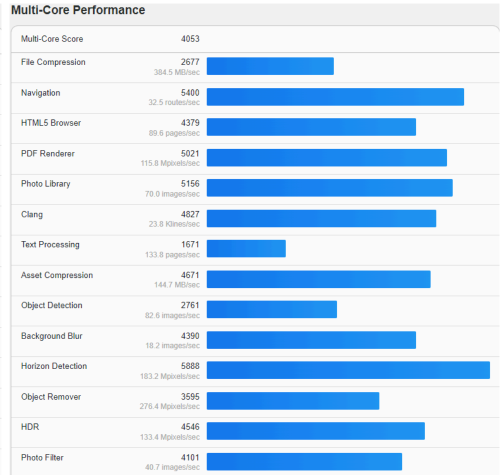
System info:



Single-core Performance:

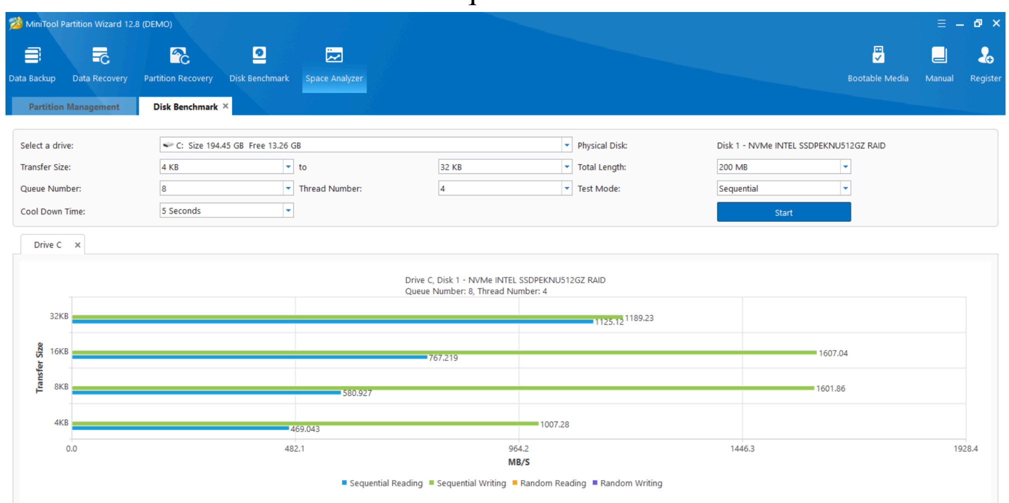


Multi-core Performance:

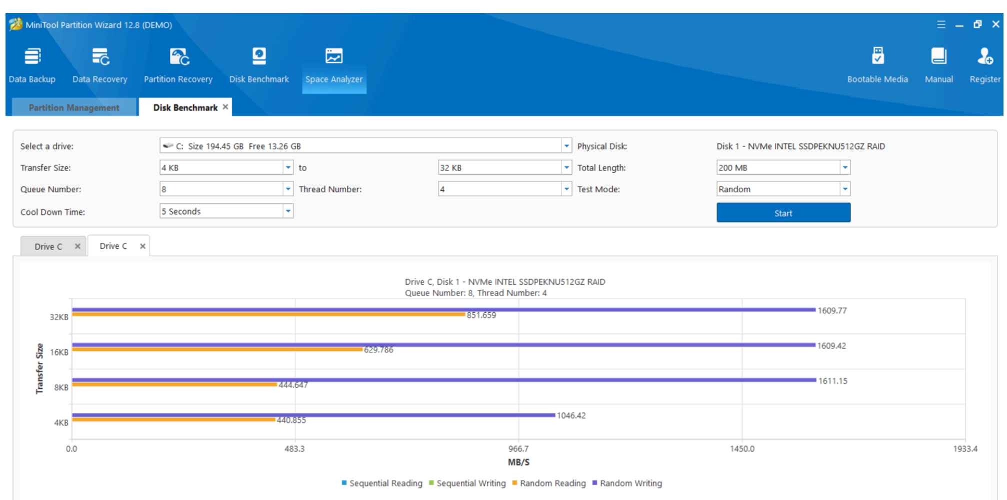


MiniTools (Disk benchmark)

Sequential:



Random:



<table>
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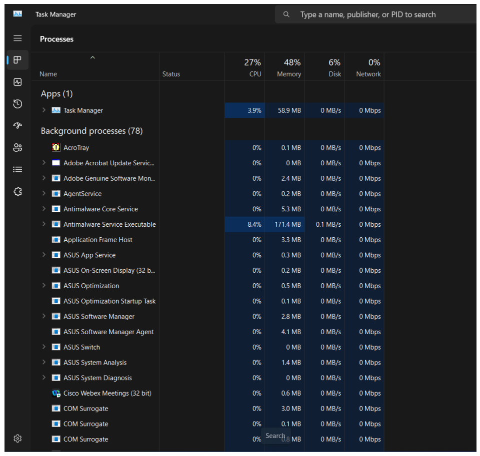
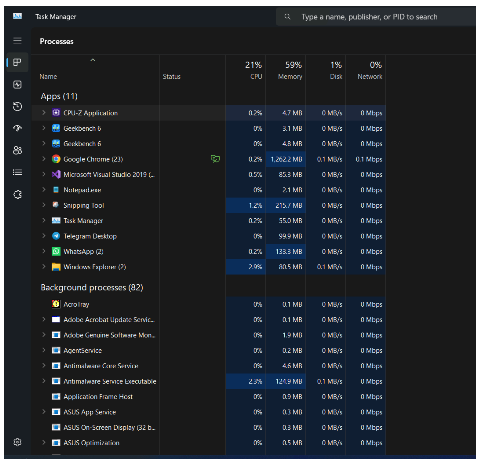
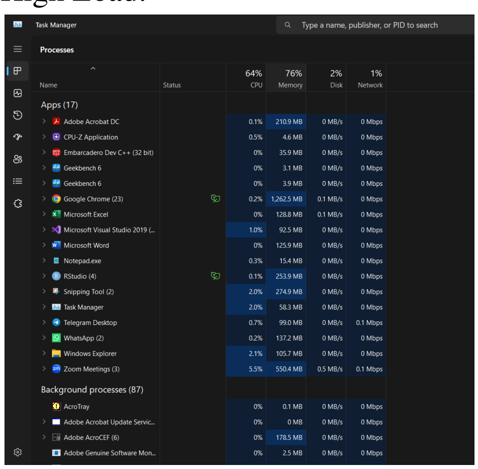
<td>Task Manager (Active Program and Resource Adjustment)</td>
<td>
Low Load:



Middle Load:



High Load:



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</tr>
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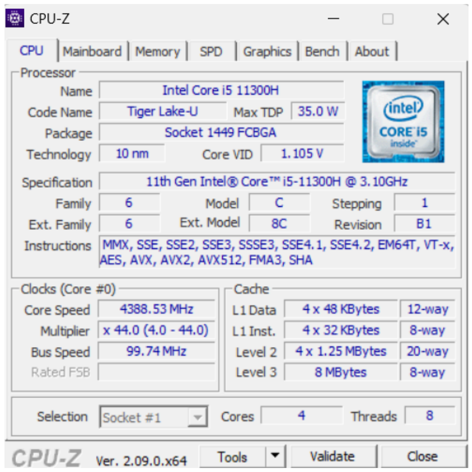
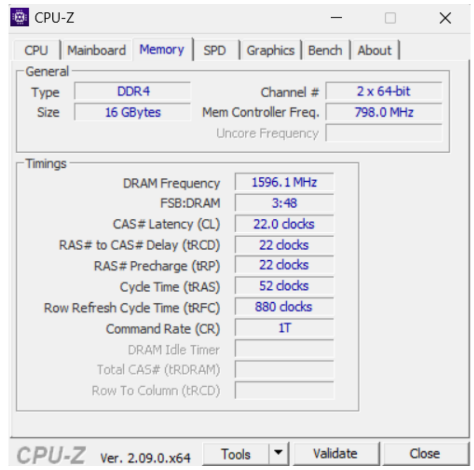
Computer B (KOO XUAN)

CPUZ (System Info)

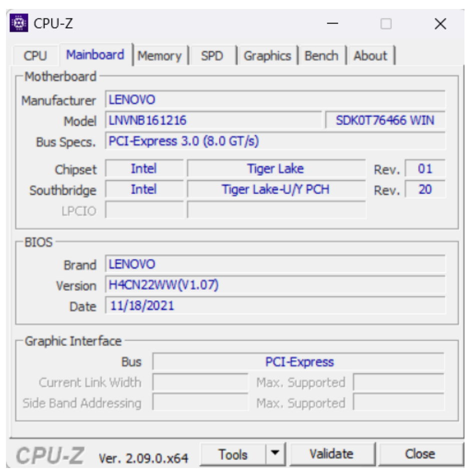
CPU Processor and cache :



Memory :

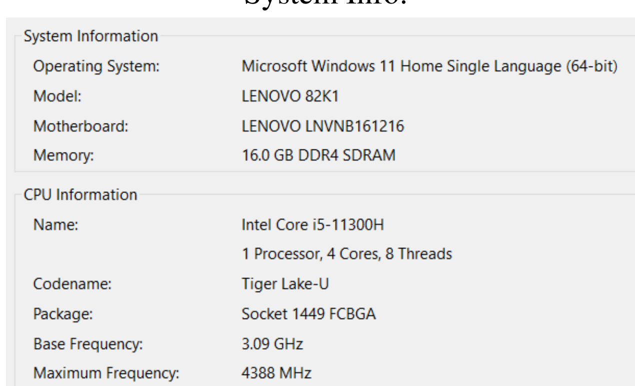


Motherboard :

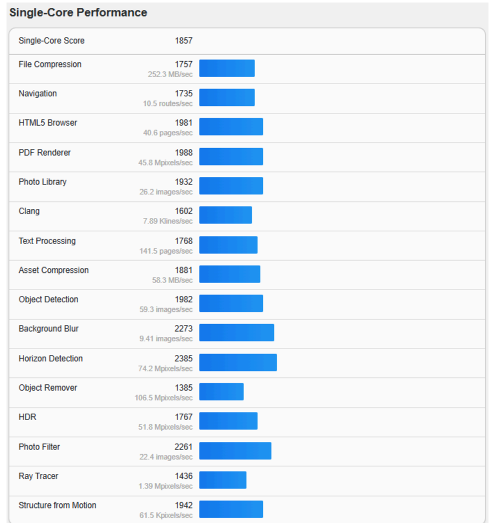


Geekbench (CPU stress test)

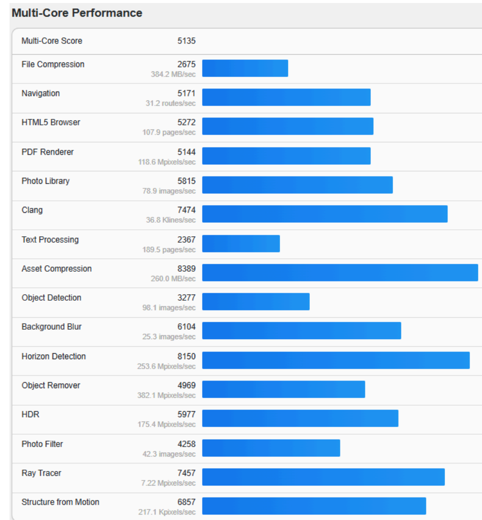
System Info:



Single-core Performance:

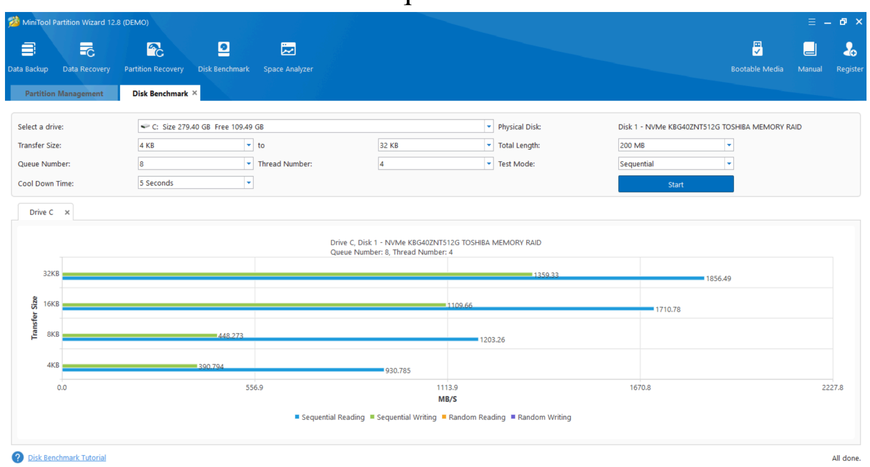


Multi-core Performance:

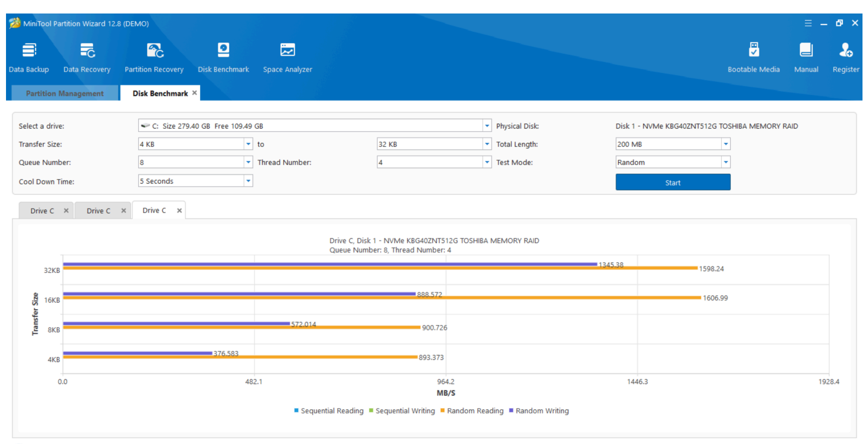


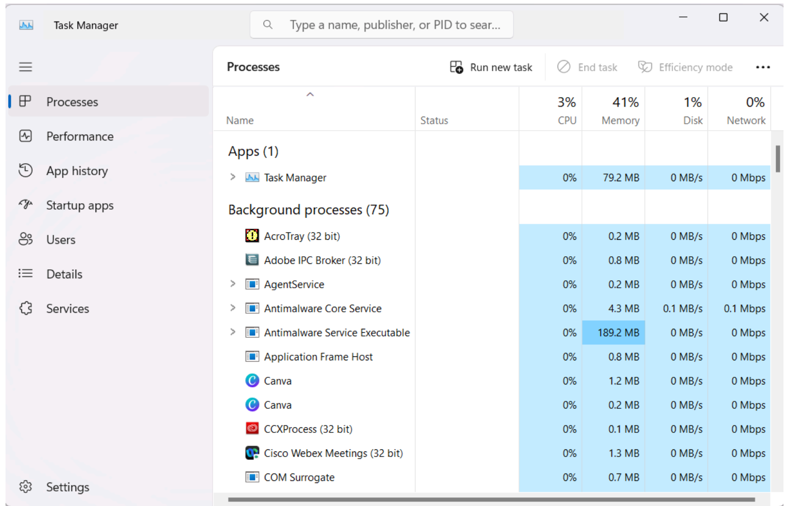
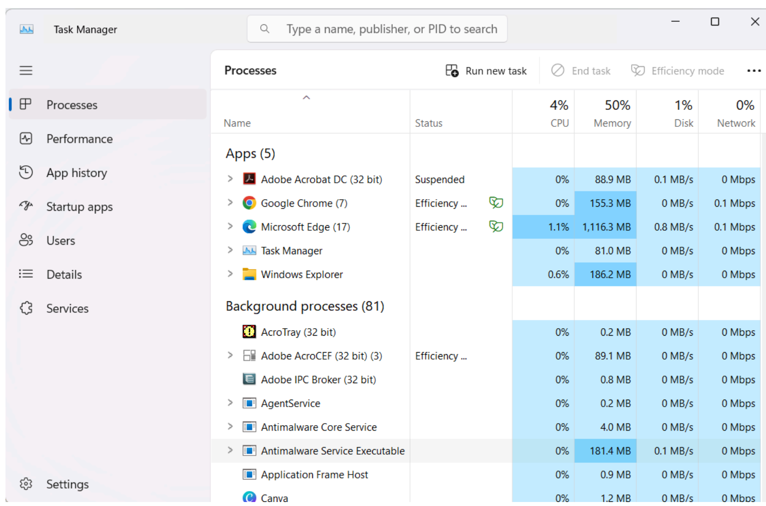
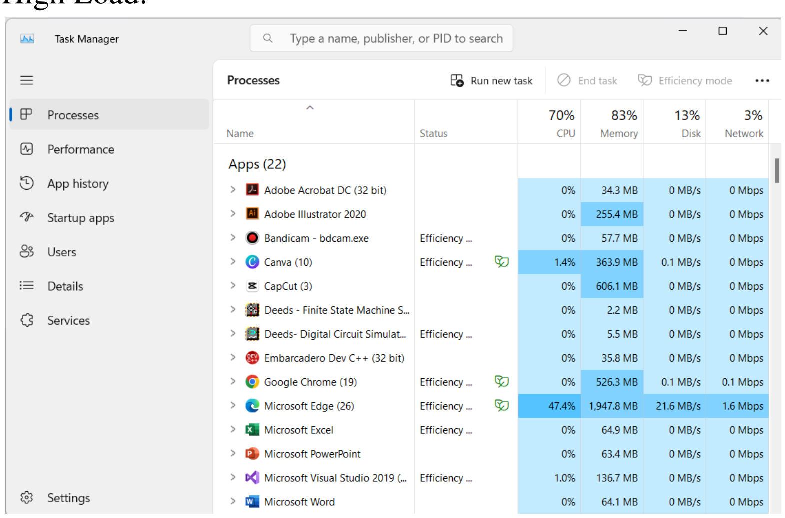
MiniTools (Disk benchmark)

Sequential:



Random:

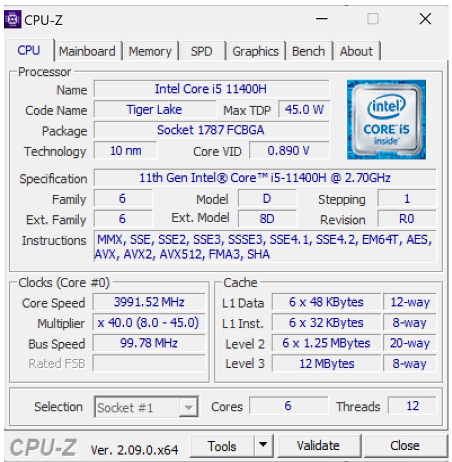


Task Manager (Active Program and Resource Adjustment)	Low Load:  Middle Load:  High Load: 

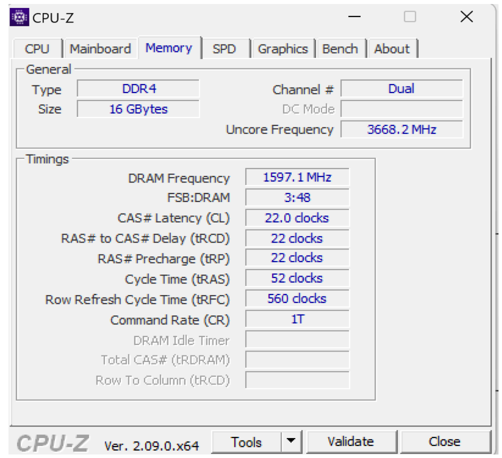
COMPUTER C (LING YU QIAN)

CPUZ
(System Info)

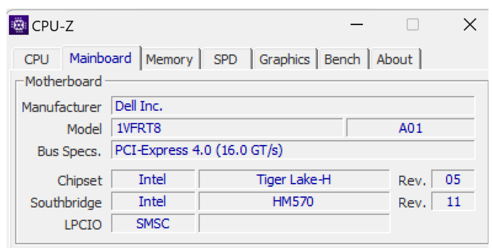
CPU Processor and cache :



memory:

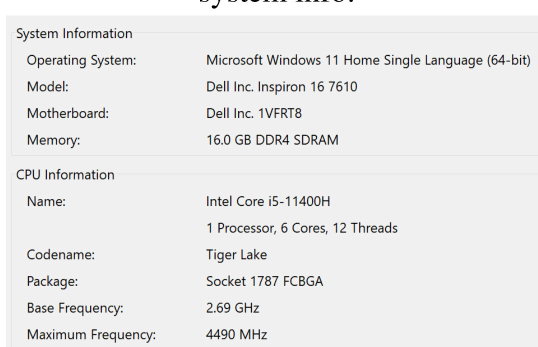


motherboard:



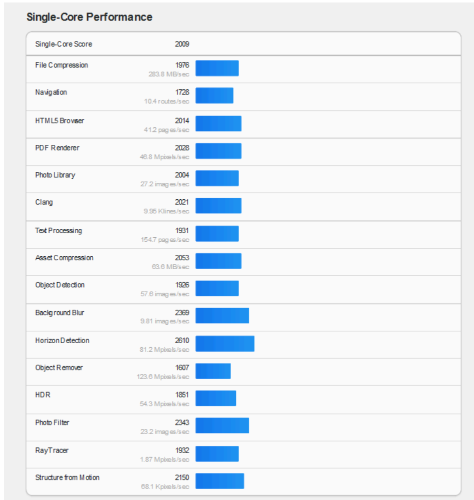
Geekbench
(CPU stress
test)

system info:

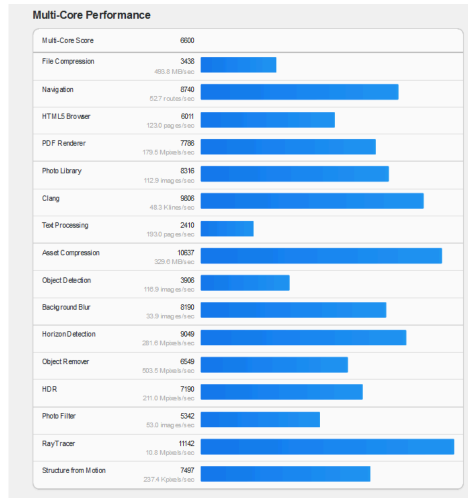


System Information	
Operating System:	Microsoft Windows 11 Home Single Language (64-bit)
Model:	Dell Inc. Inspiron 16 7610
Motherboard:	Dell Inc. 1VFRT8
Memory:	16.0 GB DDR4 SDRAM
CPU Information	
Name:	Intel Core i5-11400H
	1 Processor, 6 Cores, 12 Threads
Codename:	Tiger Lake
Package:	Socket 1787 FCBGA
Base Frequency:	2.69 GHz
Maximum Frequency:	4490 MHz

Single-core Performance:

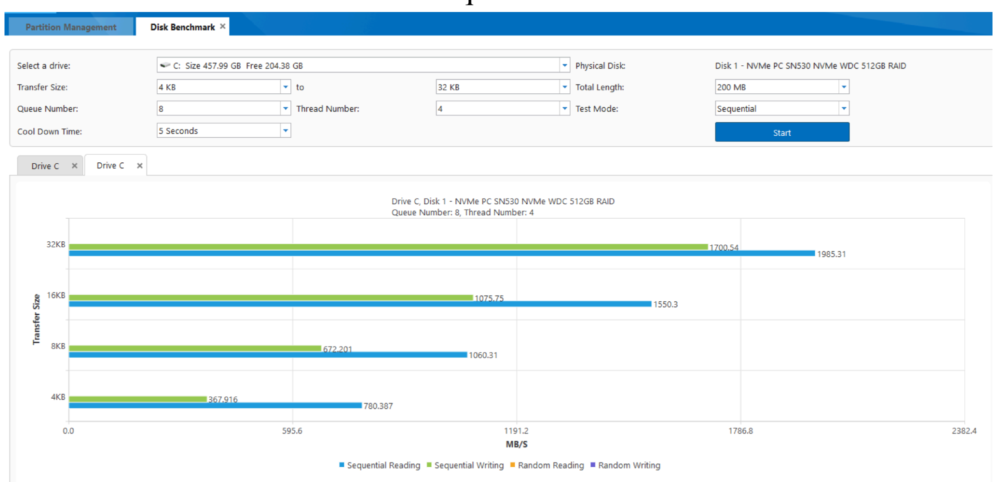


Multi-core Performance:

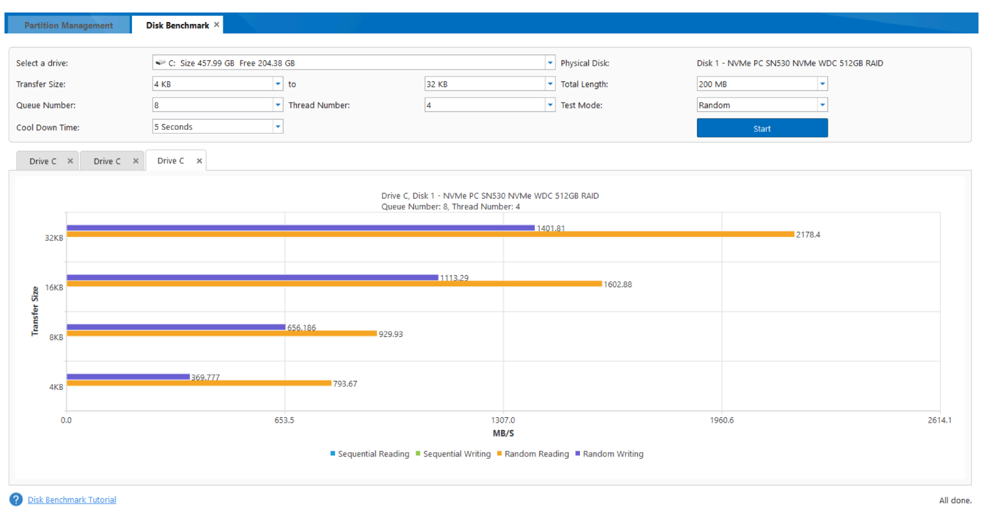


MiniTools (Disk benchmark)

Sequential:

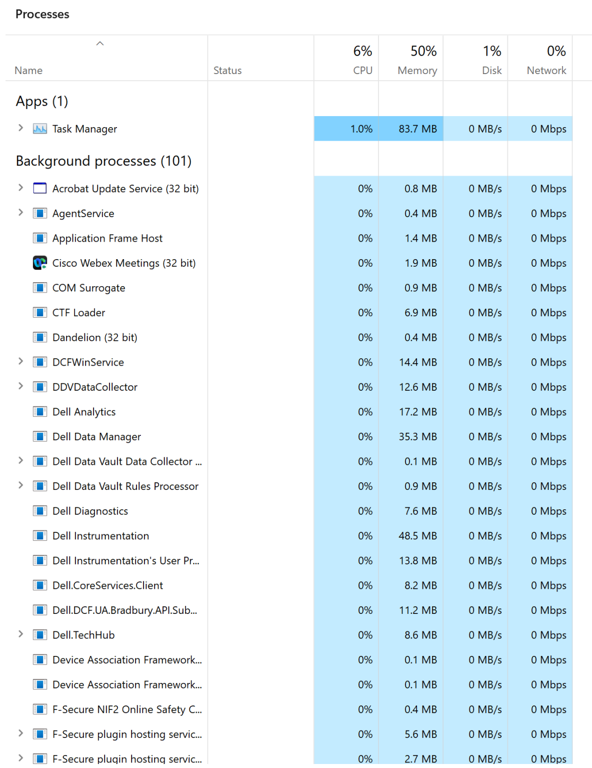


Random:



Task Manager (Active Program and Resource Adjustment)

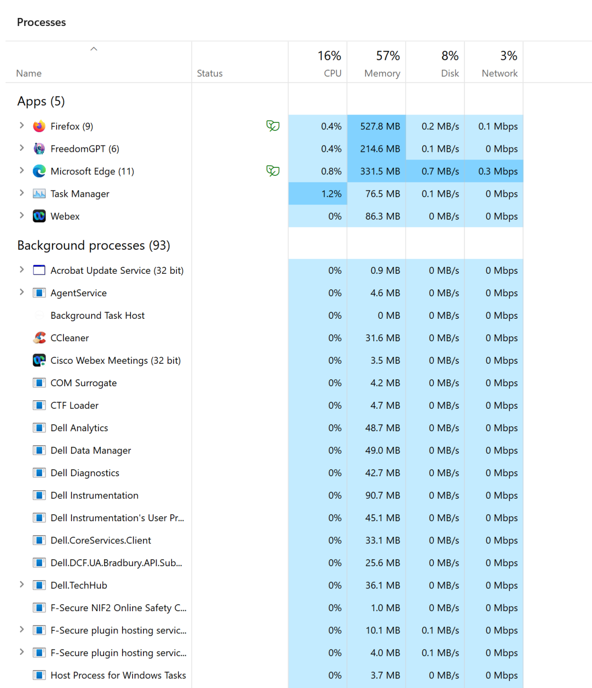
Low Load:



Processes

Name	Status	6% CPU	50% Memory	1% Disk	0% Network
Apps (1)					
Task Manager		1.0%	83.7 MB	0 MB/s	0 Mbps
Background processes (101)					
Acrobat Update Service (32 bit)		0%	0.8 MB	0 MB/s	0 Mbps
AgentService		0%	0.4 MB	0 MB/s	0 Mbps
Application Frame Host		0%	1.4 MB	0 MB/s	0 Mbps
Cisco Webex Meetings (32 bit)		0%	1.9 MB	0 MB/s	0 Mbps
COM Surrogate		0%	0.9 MB	0 MB/s	0 Mbps
CTF Loader		0%	6.9 MB	0 MB/s	0 Mbps
Dandelion (32 bit)		0%	0.4 MB	0 MB/s	0 Mbps
DCFWinService		0%	14.4 MB	0 MB/s	0 Mbps
DDVDataCollector		0%	12.6 MB	0 MB/s	0 Mbps
Dell Analytics		0%	17.2 MB	0 MB/s	0 Mbps
Dell Data Manager		0%	35.3 MB	0 MB/s	0 Mbps
Dell Data Vault Data Collector ...		0%	0.1 MB	0 MB/s	0 Mbps
Dell Data Vault Rules Processor		0%	0.9 MB	0 MB/s	0 Mbps
Dell Diagnostics		0%	7.6 MB	0 MB/s	0 Mbps
Dell Instrumentation		0%	48.5 MB	0 MB/s	0 Mbps
Dell Instrumentation's User Pr...		0%	13.8 MB	0 MB/s	0 Mbps
Dell.CoreServices.Client		0%	8.2 MB	0 MB/s	0 Mbps
Dell.DCF.UA.Bradbury.API.Sub...		0%	11.2 MB	0 MB/s	0 Mbps
Dell.TechHub		0%	8.6 MB	0 MB/s	0 Mbps
Device Association Framework...		0%	0.1 MB	0 MB/s	0 Mbps
Device Association Framework...		0%	0.1 MB	0 MB/s	0 Mbps
F-Secure NIF2 Online Safety C...		0%	0.4 MB	0 MB/s	0 Mbps
F-Secure plugin hosting servic...		0%	5.6 MB	0 MB/s	0 Mbps
F-Secure plugin hosting servic...		0%	2.7 MB	0 MB/s	0 Mbps

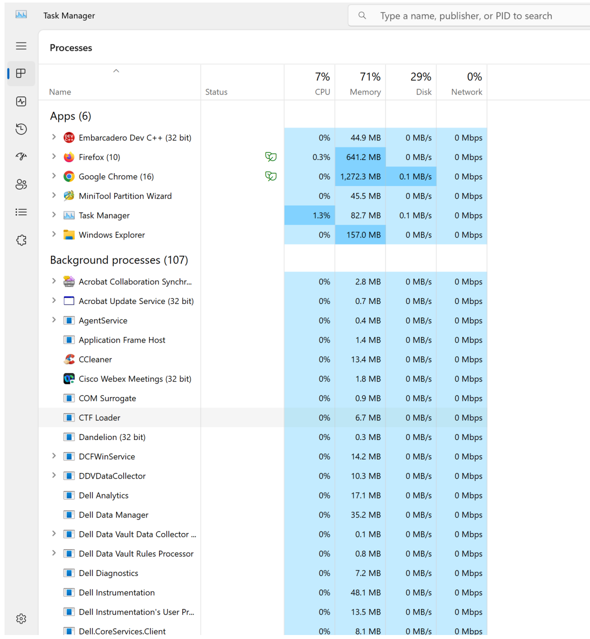
Middle Load:



Processes

Name	Status	16% CPU	57% Memory	8% Disk	3% Network
Apps (5)					
Firefox (9)		0.4%	527.8 MB	0.2 MB/s	0.1 Mbps
FreedomGPT (6)		0.4%	214.6 MB	0.1 MB/s	0 Mbps
Microsoft Edge (11)		0.8%	331.5 MB	0.7 MB/s	0.3 Mbps
Task Manager		1.2%	76.5 MB	0.1 MB/s	0 Mbps
Webex		0%	86.3 MB	0 MB/s	0 Mbps
Background processes (93)					
Acrobat Update Service (32 bit)		0%	0.9 MB	0 MB/s	0 Mbps
AgentService		0%	4.6 MB	0 MB/s	0 Mbps
Background Task Host		0%	0 MB	0 MB/s	0 Mbps
CCleaner		0%	31.6 MB	0 MB/s	0 Mbps
Cisco Webex Meetings (32 bit)		0%	3.5 MB	0 MB/s	0 Mbps
COM Surrogate		0%	4.2 MB	0 MB/s	0 Mbps
CTF Loader		0%	4.7 MB	0 MB/s	0 Mbps
Dell Analytics		0%	48.7 MB	0 MB/s	0 Mbps
Dell Data Manager		0%	49.0 MB	0 MB/s	0 Mbps
Dell Diagnostics		0%	42.7 MB	0 MB/s	0 Mbps
Dell Instrumentation		0%	90.7 MB	0 MB/s	0 Mbps
Dell Instrumentation's User Pr...		0%	45.1 MB	0 MB/s	0 Mbps
Dell.CoreServices.Client		0%	33.1 MB	0 MB/s	0 Mbps
Dell.DCF.UA.Bradbury.API.Sub...		0%	25.6 MB	0 MB/s	0 Mbps
Dell.TechHub		0%	36.1 MB	0 MB/s	0 Mbps
F-Secure NIF2 Online Safety C...		0%	1.0 MB	0 MB/s	0 Mbps
F-Secure plugin hosting servic...		0%	10.1 MB	0.1 MB/s	0 Mbps
F-Secure plugin hosting servic...		0%	4.0 MB	0.1 MB/s	0 Mbps
Host Process for Windows Tasks		0%	3.7 MB	0 MB/s	0 Mbps
Host Process for Windows Tasks		0%	0.9 MB	0 MB/s	0 Mbps

High Load:


Task Manager
Type a name, publisher, or PID to search
Processes
7% CPU
71% Memory
29% Disk
0% Network
Name
Status
Apps (6)
Embarcadero Dev C++ (32 bit) 0% 44.9 MB 0 MB/s 0 Mbps
Firefox (10) 0.3% 641.2 MB 0 MB/s 0 Mbps
Google Chrome (16) 0% 1,272.3 MB 0.1 MB/s 0 Mbps
MiniTool Partition Wizard 0% 45.5 MB 0 MB/s 0 Mbps
Task Manager 1.3% 82.7 MB 0.1 MB/s 0 Mbps
Windows Explorer 0% 157.0 MB 0 MB/s 0 Mbps
Background processes (107)
Acrobat Collaboration Synchr... 0% 2.8 MB 0 MB/s 0 Mbps
Acrobat Update Service (32 bit) 0% 0.7 MB 0 MB/s 0 Mbps
AgentService 0% 0.4 MB 0 MB/s 0 Mbps
Application Frame Host 0% 1.4 MB 0 MB/s 0 Mbps
CCleaner 0% 13.4 MB 0 MB/s 0 Mbps
Cisco Webex Meetings (32 bit) 0% 1.8 MB 0 MB/s 0 Mbps
COM Surrogate 0% 0.9 MB 0 MB/s 0 Mbps
CTF Loader 0% 6.7 MB 0 MB/s 0 Mbps
Dandelion (32 bit) 0% 0.3 MB 0 MB/s 0 Mbps
DCFWinService 0% 14.2 MB 0 MB/s 0 Mbps
DDVDataCollector 0% 10.3 MB 0 MB/s 0 Mbps
Dell Analytics 0% 17.1 MB 0 MB/s 0 Mbps
Dell Data Manager 0% 35.2 MB 0 MB/s 0 Mbps
Dell Data Vault Data Collector ... 0% 0.1 MB 0 MB/s 0 Mbps
Dell Data Vault Rules Processor 0% 0.8 MB 0 MB/s 0 Mbps
Dell Diagnostics 0% 7.2 MB 0 MB/s 0 Mbps
Dell Instrumentation 0% 48.1 MB 0 MB/s 0 Mbps
Dell Instrumentation's User Pr... 0% 13.5 MB 0 MB/s 0 Mbps
Dell.CoreServices.Client 0% 8.1 MB 0 MB/s 0 Mbps

2.4 Analysis, Comparison, Discussion

2.4.1 CPU-Z

Processor

Aspect	Computer type		
	Asus VivoBook	Lenovo 82K1	Dell Inspiron 16
Name	Intel Core i5 1135G7	Intel Core i5 11300H	Intel Core i5 11400H
Code Name	Tiger Lake-U	Tiger Lake-U	Tiger Lake
Package	Socket 1449 FCBGA	Socket 1449 FCBGA	Socket 1787 FCBGA
Technology	10nm	10nm	10nm
Specification	11th Gen Intel® Core ™ i5-1135G7	11th Gen Intel® Core ™ i5-11300H	11th Gen Intel® Core ™ i5-11400H
Stepping	1	1	1
Revision	B1	B1	R0

Based on the CPU-Z on processor aspects, all the three laptops use 11th Gen Intel® Core ™ i5 processors based on Tiger Lake architecture with 10nm technology, ensuring a good balance of performance and power efficiency. Asus VivoBook and Lenovo 82K1 uses the same socket type which is Socket 1449 FCBGA, while Dell Inspiron 16 uses Socket 1787 FCBGA, indicating the different motherboard compatibilities. The stepping of all processors are the same which is 1, however the revision differs, with Asus VivoBook and Lenovo 82K1 at B1 and Dell Inspiron 16 at R0. This difference in the revision might affect the performance stability and power efficiency.

Clocks(Core #0)

Aspect	Computer type		
	Asus VivoBook	Lenovo 82K1	Dell Inspiron 16
Core Speed	2297.10 MHz	4388.53 MHz	3991.5 MHz
Multiplier	x 23.0 (4.0 - 42.0)	x 44.0 (4.0 - 44.0)	x40.0 (8.0 - 45.0)
Bus Speed	99.79 MHz	99.74 MHz	99.78 MHz
Cores	4	4	6
Threads	8	8	12

From the comparison above, we can see that Lenovo 82K1 has a core speed of 4388.53 MHz which is significantly higher than Asus VivoBook of 2297.10MHz and Dell Inspiron 16 of 3991.5MHz. The higher multiplier of Lenovo 82K1 with x44.0 will further enhance its performance capabilities under maximum load, making it an excellent choice for tasks requiring high-single threaded performance. Dell Inspiron 16 with a core speed of 3991.5MHz and multiplier x40.0 also provided a strong performance but is optimized for multi-threaded applications due to its higher core and thread count. Asus VivoBook with a core speed of 2297.10MHz and multiplier of x23.0 is the least powerful among these three computers but it is still adequate for everyday computing tasks. The bus speed across all three laptops is nearly identical, all over around 99.7MHz, indicating similar data transfer capabilities. Moreover, Dell Inspiron 16 stands out with 6 cores and 12 threads, making it superior for multi-threaded applications and heavy multitasking. In contrast, both Asus VivoBook and Lenovo 82K1 have only 4 cores and 8 threads, which are suitable for general computing tasks.

Cache

Aspect	Computer type		
	Asus VivoBook	**Lenovo 82K1**	**Dell Inspiron 16**
L1 Data	**4 x 48 KBytes (12-way)**	**4 x 48 KBytes (12-way)**	**6 x 48 KBytes (12-way)**
L1 Inst.	**4 x 32 KBytes (8-way)**	**4 x 32 KBytes (8-way)**	**6 x 32 KBytes (8-way)**
Level 2	**4 x 1.25 MBytes (20-way)**	**4 x 1.25 MBytes (20-way)**	**6 x 1.25 MBytes (20-way)**
Level 3	**8 MBytes (8-way)**	**8 MBytes (8-way)**	**12 MBytes (8-way)**

For comparison of cache between laptops, Asus VivoBook and Lenovo 82K1 have 4x48 KBytes caches for L1 Data Cache configured in a 12-way associative setup, whereas Dell Inspiron 16 has a slightly larger cache with 6x48KBytes, also in 12-way setup. Regarding the L1 Instruction Cache, Asus VivoBook and Lenovo 82K1 both have 4x32 KBytes in an 8-way associative configuration, while Dell Inspiron 16 has 6x32 KBytes in the same 8-way setup. The Level 2 Cache is identical for Asus VivoBook and Lenovo 82K1, with each 4x1.25 MBytes in 20-way associative setup. Dell Inspiron 16 has a larger cache at this level, with 6x1.25 MBytes and also in 20-way configuration. For Level 3 Cache, all the three laptops have the same 8-way configuration however Asus VivoBook and Lenovo have a smaller feature of 8MBytes compared to Dell INspiron 16 with a larger 12 MBytes cache. Overall, Dell Inspiron 16 generally offers more cache at each level compared to Asus VivoBook and Lenovo 82K1, which have identical cache configurations.

Memory

General

Aspect	Computer type		
	Asus VivoBook	Lenovo 82K1	Dell Inspiron 16
Type	DDR 4	DDR 4	DDR 4
Size	16 GBytes	16 GBytes	16 GBytes
Channel #	2 x 64-bit	2 x 64-bit	Dual
Uncore Frequency	-	-	3668.2 MHz
Mem Controller Freq.	797.2 MHz	798.0 MHz	-

Based on the table above, we can notice that the general memory specifications are similar for all the three laptops. All the three computers use DDR 4 memory with a size of 16 GBytes. For memory channels, Asus VivoBook and Lenovo 82K1 both have 2x64-bit memory channels, whereas Dell Inspiron 16 is noted to have a Dual channel configuration, which typically implies a dual-channel setup. Dell Inspiron 16 has an Uncore Frequency of 3668.2MHz however this parameter is not specified for Asus VivoBook and Lenovo 82K1 laptops. The memory controller frequency is provided for the Asus VivoBook at 797.2MHz and Lenovo 82K1 at 798.0MHz, but this detail is absent for Dell Inspiron 16.

Timings

Aspect	Computer type		
	Asus VivoBook	**Lenovo 82K1**	**Dell Inspiron 16**
DRAM Frequency	**1600.2 MHz**	**1596.1 MHz**	**1597.1 MHz**
FSB:DRAM	**3:48**	**3:48**	**3:48**
CAS # Latency(CL)	**22.0 clocks**	**22.0 clocks**	**22.0 clocks**
RAS # to CAS # Delay (tRCD)	**22 clocks**	**22 clocks**	**22 clocks**
RAS # Precharge (tRP)	**22 clocks**	**22 clocks**	**22 clocks**
Cycle Time (tRAS)	**52 clocks**	**52 clocks**	**52 clocks**
Row Refresh Cycle Time (tRFC)	**560 clocks**	**880 clocks**	**560 clocks**
Command Rate (CR)	**1T**	**1T**	**1T**

For memory timings, we just focus on the DRAM frequency for three laptop types. The Asus VivoBook operates at a DRAM frequency of 1600.2MHz, while Lenovo 82K1 has slightly lower at 1596.1MHz and for Dell Inspiron 16 is at 1597.1MHz. These frequencies indicate that while all three models use DDR 4 memory, there are slight variations in their operating speeds. Asus VivoBook has the highest DRAM frequency among three computers, suggesting potentially better memory performance although the differences are minimal.

Motherboard

Aspect	Computer type		
	Asus VivoBook	**Lenovo 82K1**	**Dell Inspiron 16**
Manufacturer	**ASUSTeK COMPUTER INC.**	**LENOVO**	**Dell Inc.**
Model	**X515EP**	**LNVNB161216**	**1VFRT8**
Bus Spec.	**PCI-Express 4.0 (16.0 GT/s)**	**PCI-Express 3.0 (8.0 GT/s)**	**PCI-Express 4.0 (16.0 GT/s)**
Chipset	**Intel, Tiger Lake**	**Intel, Tiger Lake**	**Intel, Tiger Lake-H**
Southbridge	**Intel, Tiger Lake-U/Y PCH**	**Intel, Tiger Lake-U/Y PCH**	**HM570**

The motherboard specifications of three computers reveal significant differences and similarities in key areas such as manufacturer, model, bus specifications, chipset, and southbridge. The motherboard of Asus VivoBook manufactured by ASUSTeK COMPUTER INC. and for motherboard of Dell Inspiron 16 is manufactured by Dell Inc., both support the PCI-Express 4.0 with a transfer rate of 16.0 GT/s, which offers a substantial advantage in performance and data transfer capabilities over the Lenovo 82K1, manufactured by LENOVO, which only supports PCI-Express 3.0 with an 8.0 GT/s transfer rate. This disparity suggests that the Asus VivoBook and Dell Inspiron 16 are better suited for tasks requiring higher bandwidth, such as advanced graphics processing and faster data transfers. Both the Asus VivoBook and Lenovo 82K1 use the Intel Tiger Lake chipset paired with the Intel Tiger Lake-U/Y PCH southbridge, indicating similar basic performance capabilities. In contrast, the Dell Inspiron 16 utilizes the Intel Tiger Lake-H chipset and HM570 southbridge, which are typically designed for higher performance and better support for more demanding applications due to the H-series chipset's capability to handle higher power components. This difference points to the Dell Inspiron 16 being potentially more powerful and efficient for intensive tasks compared to Asus VivoBook and Lenovo 82K1. Overall, while the three laptops are equipped with Intel's Tiger Lake architecture, the Asus VivoBook and Dell Inspiron 16 stand out with their superior bus specifications, and in the case of the Dell Inspiron 16, a more robust chipset and southbridge configuration, making them more future-proof and capable of handling demanding applications.

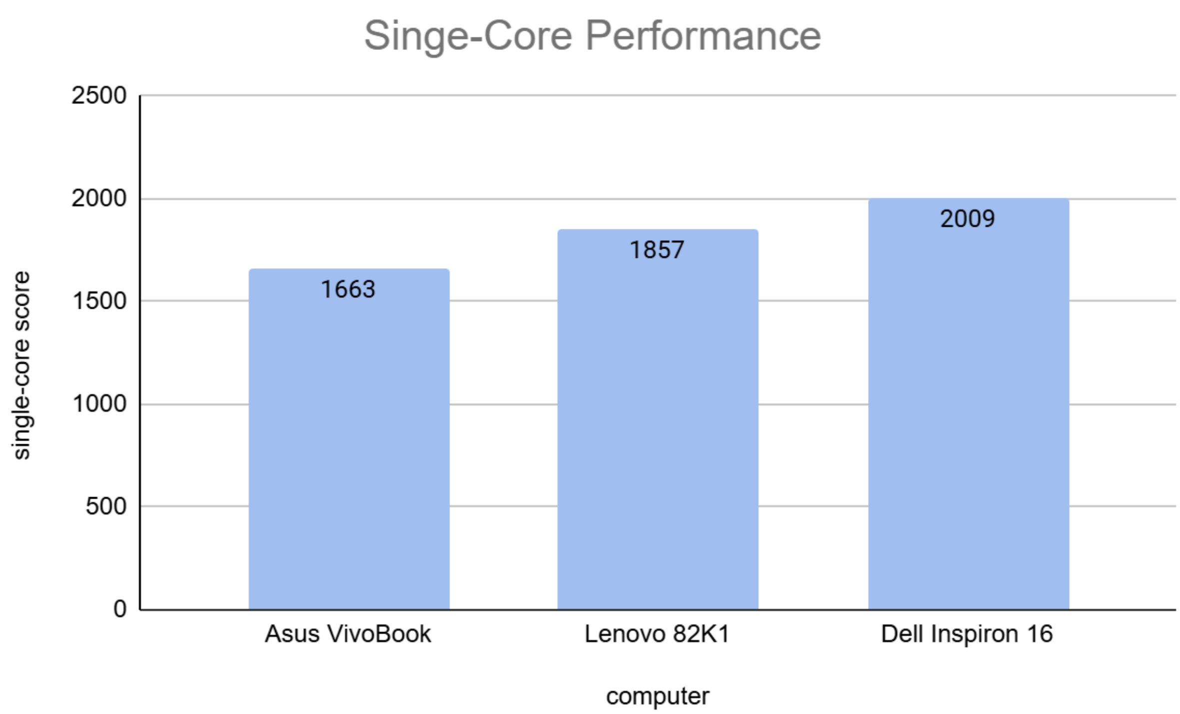
2.4.2 Geekbench

<u>COMPARISON</u>

Aspect	Computer type		
	Asus VivoBook	Lenovo 82K1	Dell Inspiron 16
SINGLE-CORE SCORE	1663	1857	2009
MULTI-CORE SCORE	4053	5135	6600
SCORE RATIO	2.44	2.77	3.29

<u>ANALYSIS</u>

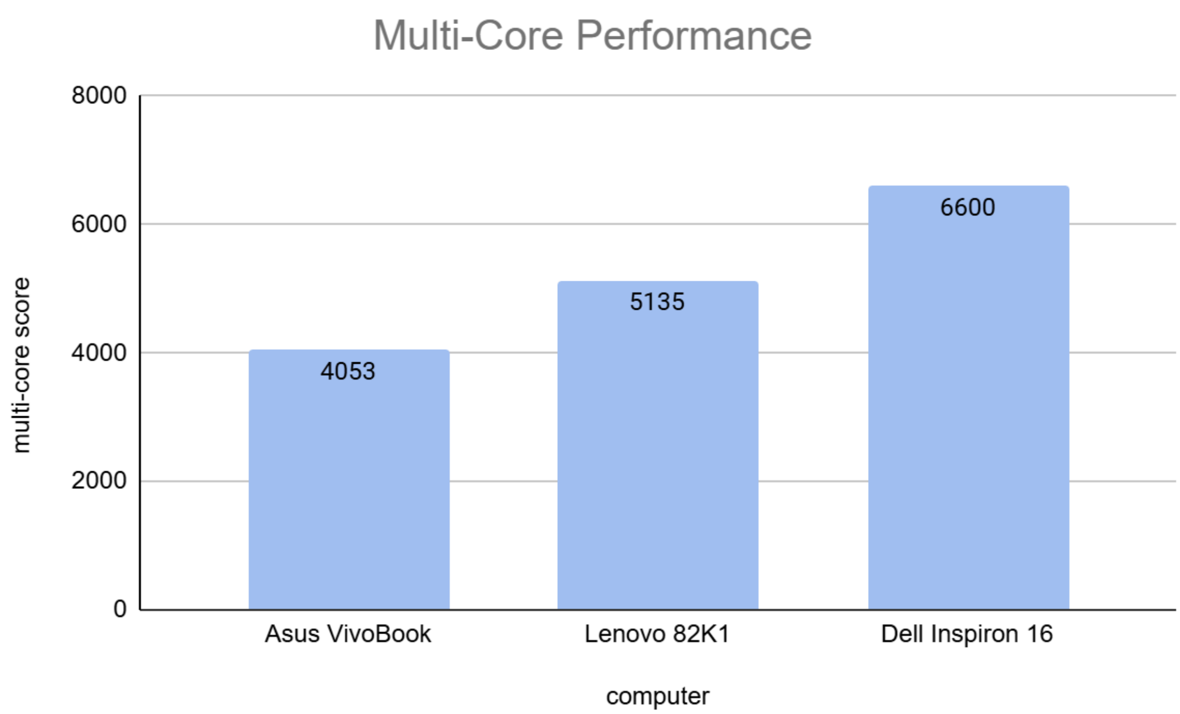
Single-Core Score



The figure above shows the comparison of the single-core performance of each CPU. From the chart above, the results show that Dell Inspiron 16 has the highest single-core score of 2009, indicating exceptional performance in tasks that rely heavily on single-threaded execution. Lenovo 82K1 follows closely with a score of 1857, demonstrating strong single-core capabilities that are likely to benefit applications such as gaming, video editing, and scientific simulations.

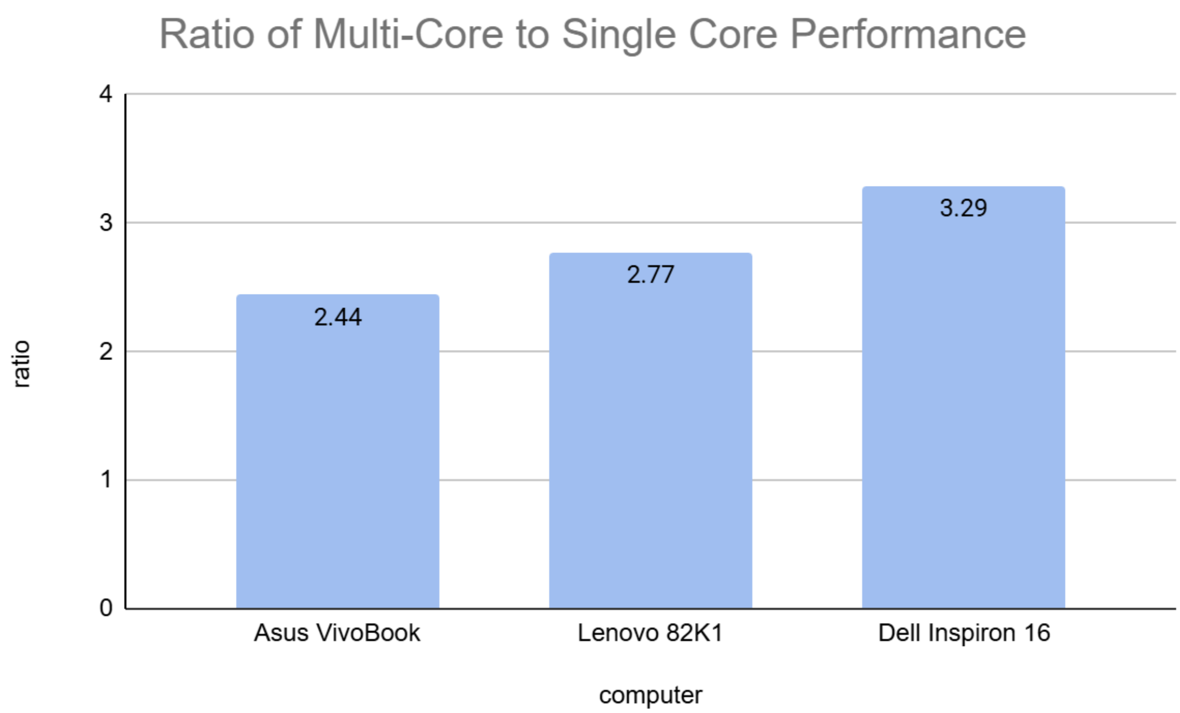
Meanwhile, AsusVivobook trails behind with a score of 1663, suggesting that it may not be as well-suited for tasks that are heavily dependent on single-core performance.

Multi-Core Score



The figure above shows the comparison of multi-core performance of each CPU. From the chart above, the results show that Dell Inspiron 16 has the highest multi-core score of 6600, indicating exceptional performance in tasks that can take advantage of multiple cores and threads. This makes it an attractive option for users who rely on heavily multi-threaded workloads, such as video encoding, 3D modeling, and scientific simulations. Lenovo 82K1 follows closely with a score of 5135, demonstrating strong multi-core capabilities suitable for applications that can utilize multiple cores. Meanwhile, Asus Vivobook trails behind with a score of 4053, suggesting that it may be less suited for tasks that require high levels of multi-core performance.

Score Ratio



The score ratios indicate that Dell Inspiron 16 excels in multi-core performance, with a ratio of 3.29, making it ideal for tasks that utilize multiple cores such as video editing or data analysis. Lenovo 82K1 has a moderate ratio of 2.77, suggesting it can handle general tasks and occasional multi-core usage. Asus Vivobook, with a ratio of 2.44, may need optimization or upgrades to improve its multi-core capabilities. In conclusion, Dell Inspiron 16 is the top choice for multi-core intensive tasks, while Lenovo 82K1 is a good option for general use, and Asus Vivobook may require further attention to reach its full potential.

DISCUSSION

The benchmarking results provide a comprehensive understanding of the performance of Asus Vivobook, Lenovo 82K1, and Dell Inspiron 16. The single-core scores indicate that Dell Inspiron 16 has the highest processing power, with a score of 2009, followed closely by Lenovo 82K1 with a score of 1857, and lastly Asus Vivobook with a score of 1663. This suggests that Dell Inspiron 16 is the most capable processor in terms of single-threaded tasks such as video editing, gaming or scientific simulation.

When it comes to multi-core performance, the results are more dramatic. Dell Inspiron 16 takes a significant lead with a score of 6600, followed by Lenovo 82K1 with a score of 5135, and then Asus Vivobook with a score of 4053. This indicates that Dell Inspiron 16 is not only strong in single-core performance but also excels in multi-core tasks, such as 3D modeling, data compression, or video encoding.

The ratio analysis provides further insights into this phenomenon. Based on the ratio analysis, Dell Inspiron 16 has the highest ratio of 3.29, indicating that it can utilize its multi-core capabilities extremely well. Lenovo 82K1 has a ratio of 2.77, which is still respectable, but not as impressive as Dell Inspiron 16's. Asus Vivobook has a ratio of 2.44, which suggests that it may require additional optimization to reach its full potential.

Overall, the benchmarking results using Geekbench indicate that Dell Inspiron 16 is the top performer in both single-core and multi-core tasks. Its exceptional performance and high ratio make it an ideal choice for a wide range of applications. Lenovo 82K1 is a solid option for users as it offers a good balance of single-core and multi-core performance. Meanwhile, Asus Vivobook may require additional optimization to reach its full potential, making it more suitable for less demanding tasks.

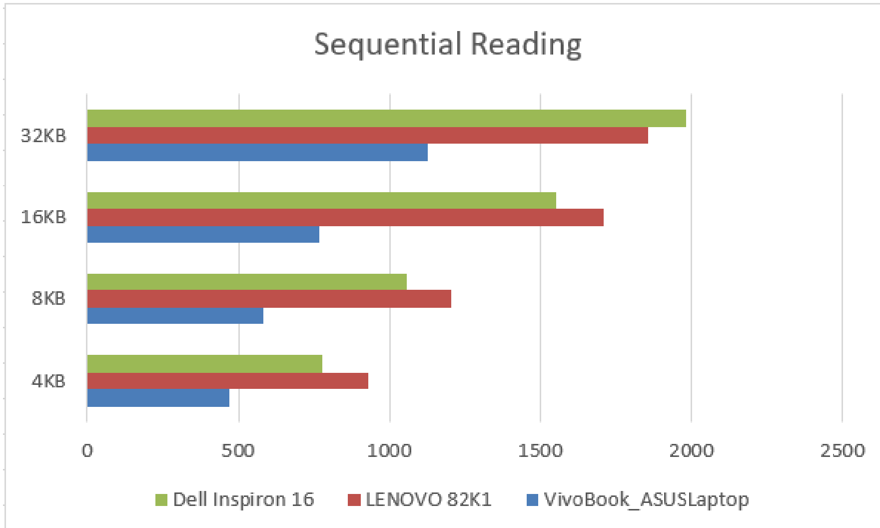
2.4.3 MiniTool

(Disk benchmark)

Stress test

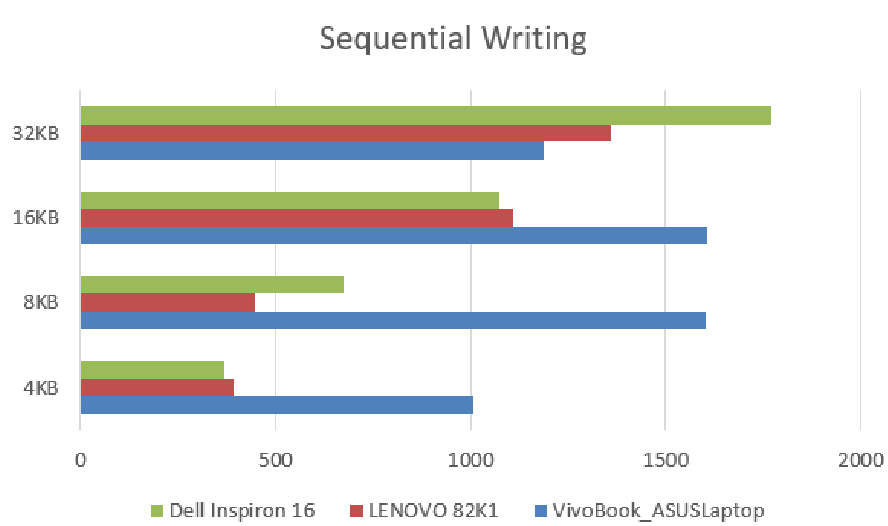
Specification				Computer Type		
				VivoBook_ASUSLaptop	LENOVO 82K1	Dell Inspiron 16
Mode test	Sequential	Reading	4KB	469.043 MB/S	930.785 MB/S	780.387 MB/S
			8KB	580.927 MB/S	1203.26 MB/S	1060.31 MB/S
			16KB	767.219 MB/S	1710.78 MB/S	1550.3 MB/S
			32KB	1125.12 MB/S	1856.49 MB/S	1985.31 MB/S
		Writing	4KB	1007.28 MB/S	390.794 MB/S	367.916 MB/S
			8KB	1601.86 MB/S	448.273 MB/S	672.201 MB/S
			16KB	1607.04 MB/S	1109.66 MB/S	1074.75 MB/S
			32KB	1189.23 MB/S	1359.33 MB/S	1770.54 MB/S
	Random	Reading	4KB	440.855 MB/S	893.373 MB/S	793.67 MB/S
			8KB	444.647 MB/S	900.726 MB/S	929.93 MB/S
			16KB	629.786 MB/S	1606.99 MB/S	1602.88 MB/S
			32KB	851.659 MB/S	1598.24 MB/S	2178.4 MB/S
		Writing	4KB	1046.42 MB/S	376.583 MB/S	369.777 MB/S
			8KB	1611.15 MB/S	572.014 MB/S	656.186 MB/S
			16KB	1609.42 MB/S	888.572 MB/S	1113.29 MB/S
			32KB	1609.77 MB/S	1345.38 MB/S	1401.81 MB/S

Sequential Reading



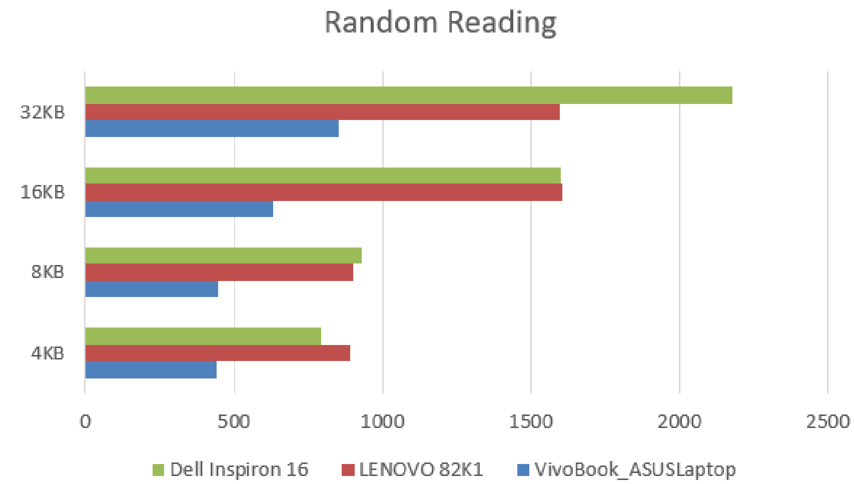
Based on the chart above, for sequential reading, we know that VivoBook_ASUSLaptop has lower performance compared to the other two systems across all transfer sizes. LENOVO 82K1 has strong performance, especially noticeable in the 16KB and 32KB transfer sizes. However, Dell Inspiron 16 has the highest performance in 32KB transfer size, indicating strong sequential read capabilities.

Sequential Writing



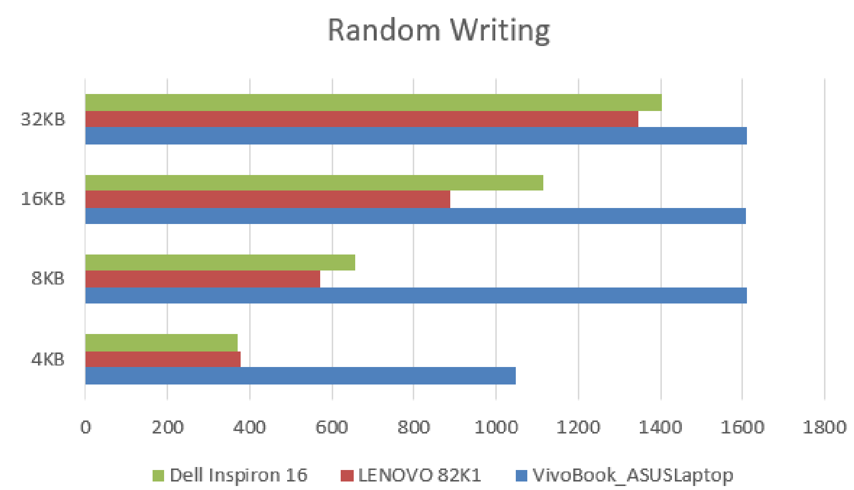
For sequential writing, VivoBook_ASUSLaptop has excellent performance at 4KB, 8KB and 16KB transfer sizes, outperforming the other systems but it is slower in higher transfer sizes at 32 KB compared to Dell Inspiron 16. Besides, LENOVO 82K1 is generally lower performance in sequential writing compared to the other systems, especially noticeable at lower transfer sizes. For Dell Inspiron 16, although it is slow at 4KB, it has strong performance at other transfer size, particularly at 32 KB transfer size.

Random Reading



For random reading, VivoBook_ASUSLaptop consistently has lower performance across all transfer sizes compared to the other systems. Whereas, LENOVO 82K1 has high performance at 16KB and 32KB transfer sizes, but slightly lower at 4KB and 8KB. In Dell Inspiron 16, it has the highest performance at 32 KB transfer size, suggesting strong random read capabilities.

Random Writing



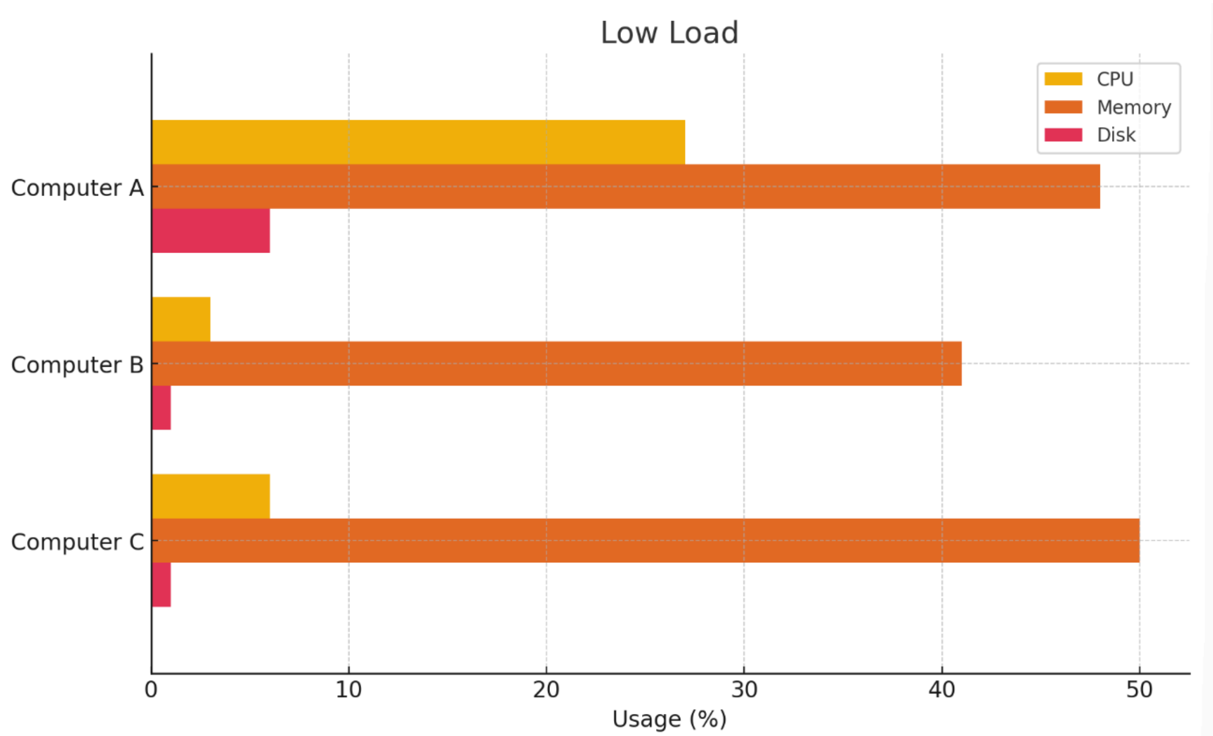
For random writing, VivoBook_ASUSLaptop has strong performance, particularly noticeable at all transfer sizes. LENOVO 82K1 has lower performance at 4KB transfer size but improves significantly at higher transfer sizes. Dell Inspiron 16 has a moderate performance across all transfer sizes, with the highest speed observed at 16KB transfer size.

By using MiniTools, this disk benchmark tool, we know that VivoBook_ASUSLaptop excels in sequential and random writing at smaller transfer sizes but lags in reading speeds. LENOVO 82K1 shows strong performance in sequential and random reading, particularly at higher transfer sizes, but has lower writing speeds at smaller transfer sizes. And Dell Inspiron 16 which is overall balanced performance with the highest sequential and random reading speeds at higher transfer sizes.

2.4.4 Task Manager

LOW LOAD

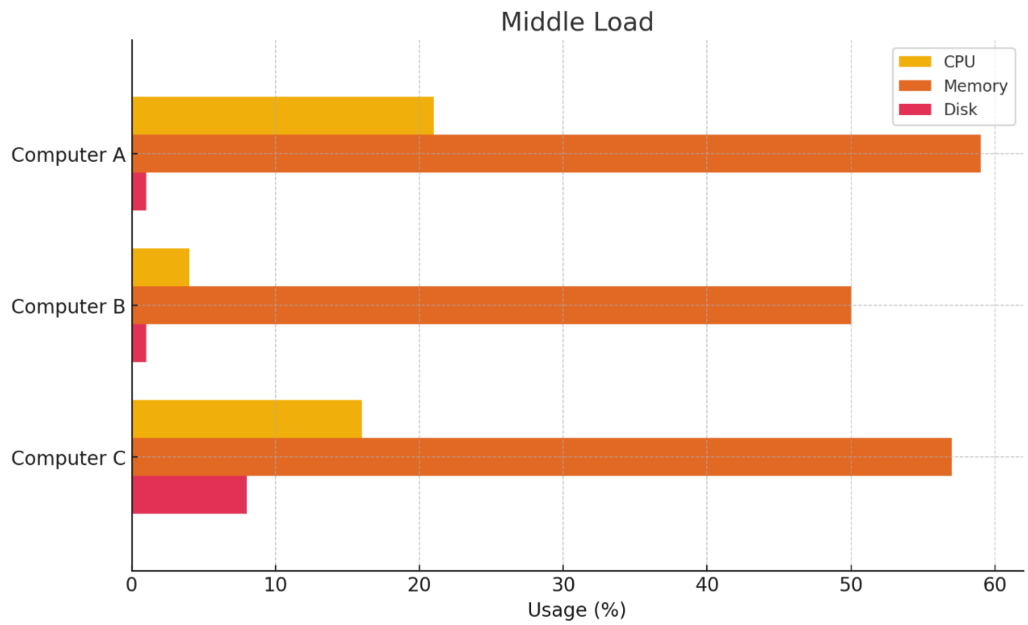
	CPU	MEMORY	DISK
computer a	27	48	6
computer b	3	41	1
computer c	6	50	1



Under low load conditions, Computer A shows a balanced performance with moderate CPU (27%) and memory (48%) usage and low disk usage (6%). This indicates that it handles tasks requiring moderate processing and memory efficiently, making it suitable for general-purpose tasks and light multitasking. Computer B exhibits very low CPU (3%) and disk (1%) usage but moderate memory usage (41%), suggesting it is well-suited for tasks that do not require significant processing power but do need a fair amount of memory. It is best for lightweight tasks, background processes, and applications with minimal processing needs. Computer C, on the other hand, shows low CPU usage (6%) and disk usage (1%) but high memory usage (50%), indicating it is engaged in memory-intensive tasks. This makes it suitable for applications that require significant memory.

MIDDLE LOAD

	CPU	MEMORY	DISK
computer a	21	59	1
computer b	4	50	1
computer c	16	57	8



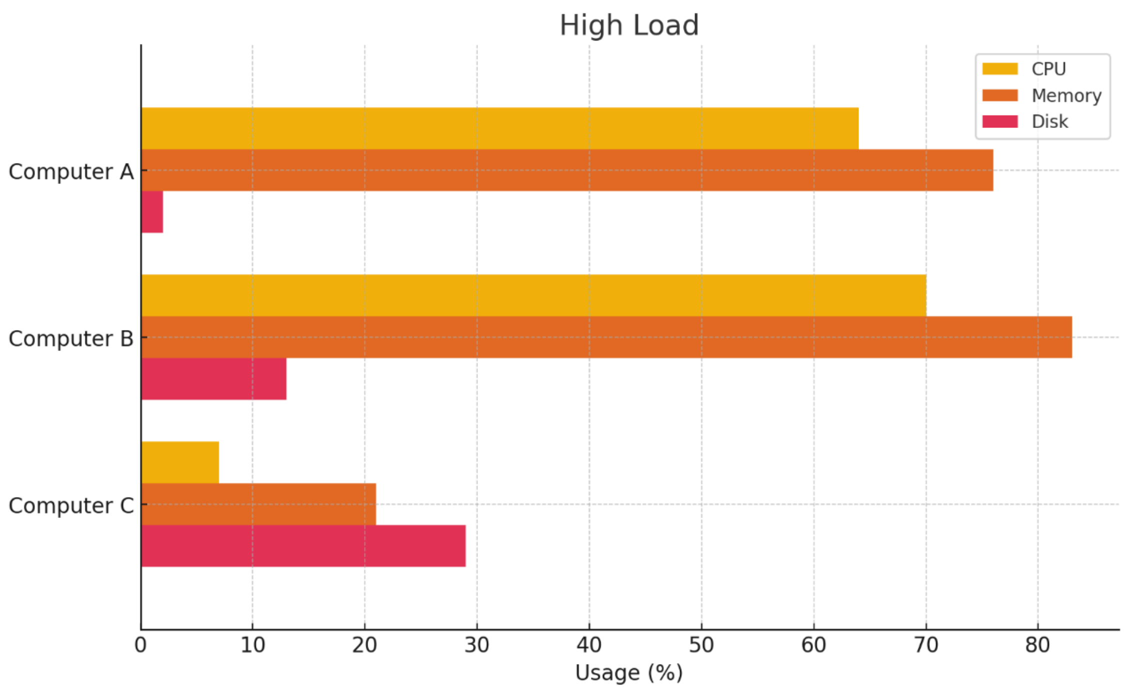
Under middle load conditions, Computer A's CPU usage decreases slightly to 21%, while its memory usage increases to 59%, and disk usage drops to 1%. This shift indicates that the system is now handling tasks that require more memory, such as more extensive multitasking or running applications with larger datasets. The minimal disk usage suggests that these tasks are still not heavily reliant on disk I/O, pointing to an efficient use of memory for caching and active data processing.

Computer B's slight increase in CPU usage to 4%, coupled with a memory usage increase to 50%, and steady low disk usage at 1%, suggests it remains balanced but leans towards handling more data in memory. This could mean the system is now engaged in more demanding tasks that still do not fully tax the CPU but require more memory, such as more complex database operations or running multiple applications simultaneously.

Computer C shows a substantial increase in memory usage to 57% and a notable increase in disk usage to 8%, while CPU usage drops to 16%. This indicates a significant shift towards handling memory and I/O-intensive tasks, likely involving more data processing and storage operations. The lower CPU usage suggests that while the tasks require substantial data handling, they are not as computationally intensive, possibly pointing to a workload involving data analysis or transfer.

HIGH LOAD

	CPU	MEMORY	DISK
computer a	64	76	2
computer b	70	83	13
computer c	7	21	29



Under high load conditions, Computer A shows high CPU (64%) and memory (76%) usage but still low disk usage at 2%. This indicates that it is engaged in intensive processing and memory-heavy applications, such as video rendering, complex simulations, or virtual machine hosting. The low disk usage suggests efficient memory and CPU utilization, with minimal reliance on disk I/O, likely due to effective caching and processing strategies.

Computer B demonstrates high usage across all metrics: CPU (70%), memory (83%), and disk (13%). This suggests it is under a comprehensive heavy load, handling tasks that demand substantial processing power, large memory capacity, and significant disk operations. This pattern is typical for systems running large-scale enterprise applications, extensive data processing tasks, or high-performance computing scenarios where all system resources are heavily utilized.

Computer C, with low CPU (7%) and memory (21%) usage but high disk usage at 29%, indicates it is primarily engaged in I/O-bound operations. This suggests tasks such as large data transfers, intensive file operations, or heavy database transactions that rely on disk performance rather than processing power or memory. This profile is indicative of a system designed for high-volume data handling and storage operations, making it suitable for roles such as a dedicated file server or a database server under heavy load.

SUMMARY

In conclusion, Computer A is best for balanced, processing, and memory-intensive tasks without heavy disk I/O. It maintains stable low disk usage, making it ideal for general-purpose tasks and light multitasking. Computer B handles lightweight tasks efficiently under low loads and scales well for more balanced tasks under higher loads, with a consistent emphasis on memory usage and low disk activity. Computer C is versatile, starting with CPU-intensive tasks under low loads and shifting towards I/O-bound tasks under higher loads, making it suitable for varied applications depending on the load condition. The key differences include Computer A's stable low disk usage, Computer B's consistent memory focus with low disk activity, and Computer C's significant shift from CPU to disk-intensive operations as load increases.

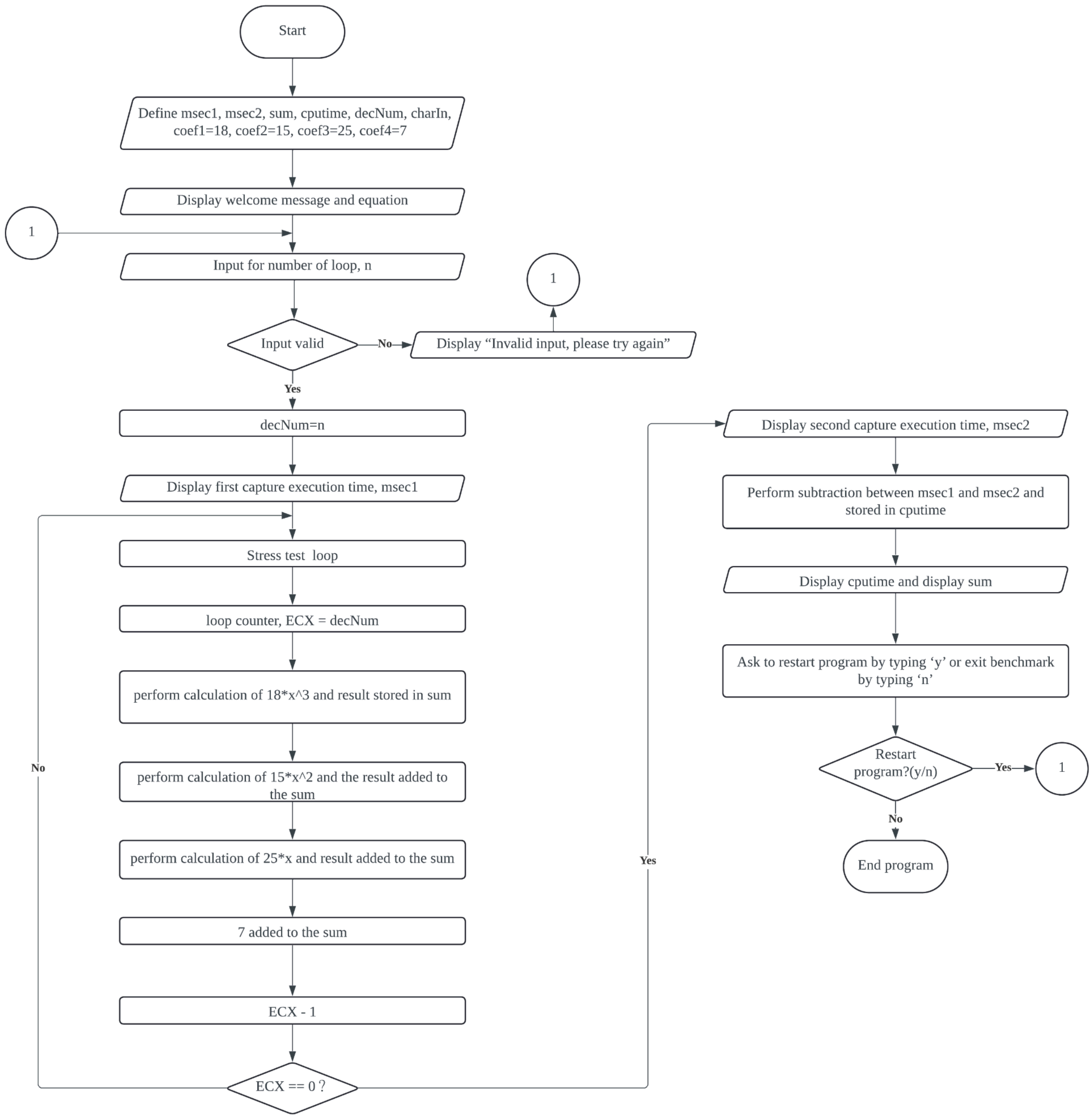
3.0 Part 2
Writing ASM code program for Benchmarking software

3.1 Overview
This part of the project is writing an assembly language program to calculate the sum and measure the time taken to execute 50, 100, 200, 500, 700 and 1000 loops for the equation:
$y = 18*x^3 + 15*x^2 + 25*x + 7$. This program would be run on 3 different computers as selected in Part 1 to compare performance differences.

This part of the project aims to show how the difference between the benchmarking criteria we interpreted above affects the execution time of real work on three different computers.

This part is divided into several sections, the description of each section is displayed below:
- flowchart of the execution: To show the algorithm of the program including the logical flow of the program's control from the beginning to the end
- coding and documentation: To embed the coding and detailed information about the coding
- execution and results: To embed the screenshots of the output after execution of the program
- analysis, comparison and discussion: To compare, analyze and discuss the output after the execution of the program between 3 computers.

3.2 Flowchart



3.3 Coding and Documentation

```
TITLE Project COA Part2
; Author:     Chin Pei Wen
;             Tan Zhao Hong
;             Koo Xuan
;             Ling Yu Qian
; Date: 4 July 2024

INCLUDE Irvine32.inc

.data
msec1 DWORD ?
msec2 DWORD ?
coef1 DWORD 18; student1 18 - 03 - 2003
coef2 DWORD 15; student2 15 - 06 - 2003
coef3 DWORD 25; student3 25 - 09 - 2004
coef4 DWORD 7; student4 07 - 12 - 2004
sum WORD ?
cputime DWORD ?
str0 BYTE "Welcome to CPU Benchmark Program", 0dh, 0ah, 0dh, 0ah, 0
str1 BYTE "Benchmark CPU time Using Equation y = 18*x^3 + 15*x^2 + 25*x +7", 0dh, 0ah
BYTE "                    (with delay coef1, coef2, voef3, coef4 = 18, 15, 25, 07 msec)",
0dh, 0ah, 0
str2 BYTE "Enter Number of Looping (N) = ", 0
str3 BYTE "Value of Sum from the Stress Test (polynomial) = ", 0
str4 BYTE "CPU time Stress Test in progress... ", 0dh, 0ah, 0
str_result BYTE "Result : ", 0dh, 0ah, 0
time1 BYTE "First Capture Execution time in milisecond: ", 0
time2 BYTE "Second Capture Execution time in milisecond: ", 0
time3 BYTE "Different Execution time in milisecond: ", 0
decNum DWORD ?
promptBad BYTE "Invalid input, please enter again", 0dh, 0ah, 0
str5 BYTE "Press 'y' to continue or 'n' to exit the benchmark: ", 0
charIn BYTE ?
charY db "y"
str6 BYTE "Thank you for using this Benchmark Program... BYE!!", 0dh, 0ah, 0

.code
main PROC
```

```
startProg :

call Clrscr
call Crlf
mov edx, OFFSET str0
call WriteString; display welcome

mov edx, OFFSET str1
call WriteString; display equation

call Crlf

mov edx, OFFSET str2
call WriteString; display no._loop

read :

call ReadDec; input no._loop
jnc goodInput

mov edx, OFFSET promptBad
call WriteString
jmp read; go input again

goodInput :

mov decNum, eax; store good value

mov edx, OFFSET str4
call WriteString; display CPU in_progress

call Crlf
mov edx, OFFSET str_result
call WriteString; display result

call Crlf

call GetMseconds
mov msec1, eax; get capture_msec before loop
```

```
lea edx, time1
call WriteString
call WriteDec

call Crlf

mov BX, 1; set x = 1
mov ecx, decNum; max_loop N

LoopStress :

mov eax, coef1; delay factor coef1
call Delay
mul bx
mul bx
mul bx
add sum, ax; partial sum for coef1

mov eax, coef2; delay factor coef2
call Delay
mul bx
mul bx
add sum, ax; partial sum for coef2

mov eax, coef3; delay factor coef3
call Delay
mul bx
add sum, ax; partial sum for coef3

mov eax, coef4; delay factor coef4
call Delay
add sum, ax; partial sum for coef4

inc bx

loop LoopStress

call GetMseconds
mov msec2, eax
```

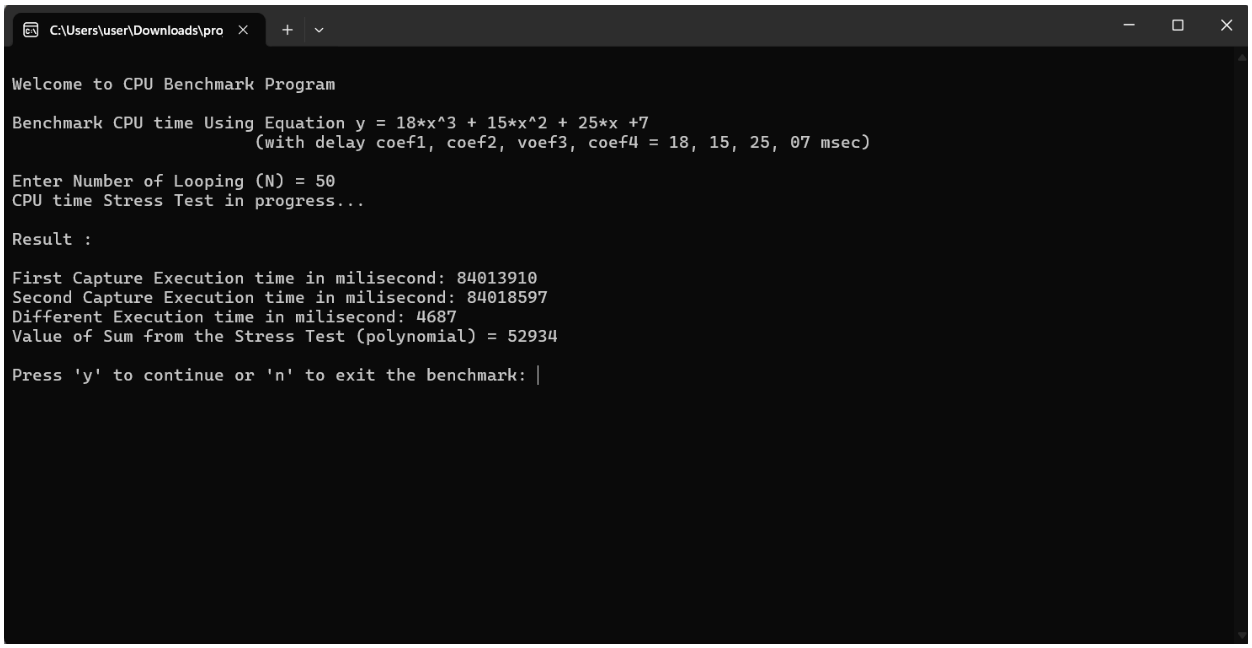
```
lea edx, time2
call WriteString
call WriteDec

call Crlf

mov eax, msec2
sub eax, msec1; calculate diff_CPUtime
mov cputime, eax

lea edx, time3
call WriteString
mov eax, cputime
call WriteDec

call Crlf
lea edx, str3
call WriteString; value sum
mov ax, sum
call WriteDec

call Crlf
call Crlf

mov edx, OFFSET str5
call WriteString; continue_or_exit

call ReadChar
mov charIn, AL

call Crlf
call Crlf

mov BL, charY
cmp BL, charIn

JE startProg

mov edx, OFFSET str6
call WriteString; thanks and bye
```

exit

main ENDP

END main

Result:

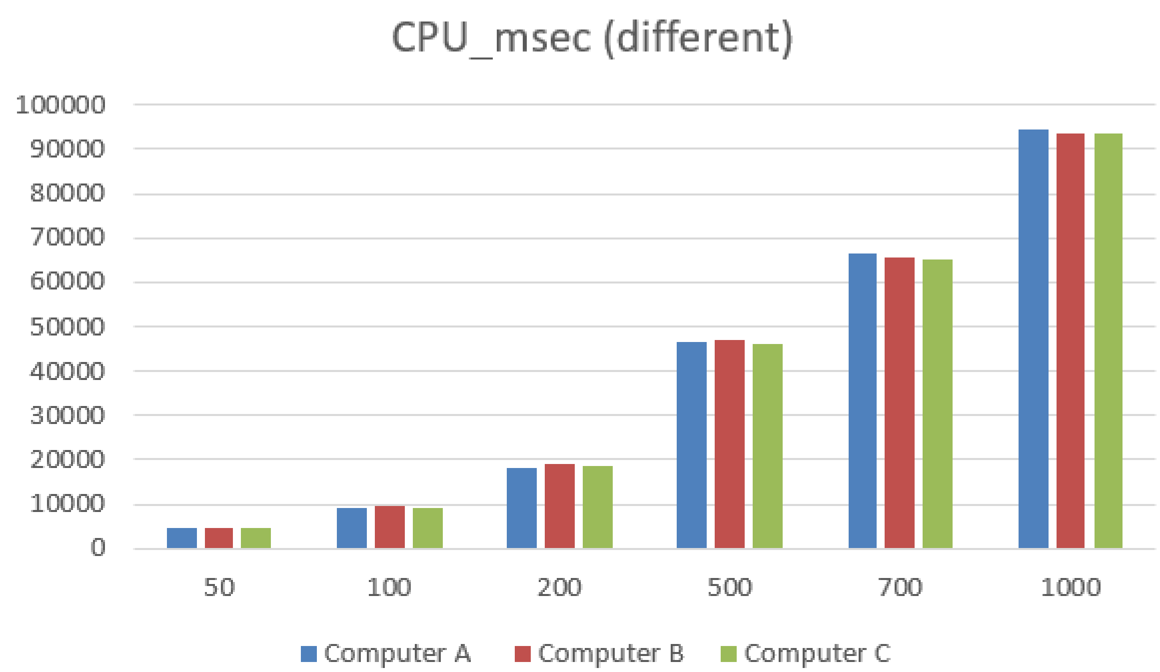


3.4 Implementation/Execution and Results

No. of Loop (N)	50	100	200	500	700	1000
Computer A						
Capture_ msec (before)	44109881	44138708	44167884	44206098	44275511	44362079
Capture_ msec (after)	44114495	44147889	44186039	44252772	44342027	44456781
CPU_msec (different)	4614	9181	18155	46674	66516	94702
Computer B						
Capture_ msec (before)	84013910	84050888	84079070	84116510	84184641	84274652
Capture_ msec (after)	84018597	84060287	84097963	84163385	84250259	84368459
CPU_msec (different)	4687	9399	18893	46875	65618	93807
Computer C						
Capture_ msec (before)	38753200	38865896	38975320	39044072	39128437	39264170
Capture_ msec (after)	38757841	38875156	38993867	39090397	39193360	39357678
CPU_msec (different)	4641	9260	18547	46325	64923	93508

3.5 Analysis, Comparison, Discussion

We have examined the CPU performance of three different computers, labeled as Computer A, Computer B, and Computer C. Our analysis focused on CPU time differences across varying loop counts to evaluate the efficiency and performance scalability of each system under increasing computational loads.



When comparing the performance of Computer A with Computer B and C, it is evident that all three systems exhibit a similar pattern in CPU time differences as the loop count increases. At 50 loops, Computer A records a CPU time difference of 4614 msec, which is the shortest runtime compared to Computer B and C. This trend continues consistently across higher loop counts. For instance, at 100 loops, Computer A takes 9181 msec, compared to Computer B's 9399 msec and Computer C's 9260 msec. At 500 loops, Computer A records 46674 msec, whereas Computer B and C take 46875 msec and 46325 msec, respectively. The pattern holds at 1000 loops, with Computer A at 94702 msec, Computer B at 93807 msec, and Computer C at 93508 msec. Overall, while Computer A performs efficiently, Computers B and C show slightly better performance, especially at higher loop counts.

Computer B demonstrates solid performance but generally has higher CPU time differences compared to Computer C, indicating slightly less efficiency. For example, at 50 loops, Computer B has a CPU time difference of 4687 ms, which is marginally higher than both Computer A (4614 ms) and Computer C (4641 ms). This pattern persists as the loop count increases at 100 loops, Computer B's CPU time difference is 9399 ms, again higher than Computer A (9181 ms) and Computer C (9260 ms). As the number of loops further increases, the differences become more pronounced, with Computer B recording 18893 ms at 200 loops and 46875 ms at 500 loops. Although its performance at 1000 loops (93807 ms) is slightly better compared to Computer A, Computer B is consistently outperformed by Computer C, especially under heavy loads. This suggests that while Computer B performs well, there is room for optimization to enhance its efficiency further.

When comparing Computer C to Computers A and B, notable differences emerge. Computer A exhibits similar trends in CPU time increases but generally shows slightly higher CPU times at greater loop counts compared to Computer C. Conversely, Computer B displays marginally higher CPU times at mid-range loops, although it aligns closely with Computer C under higher loads.In discussion, Computer C's performance across varying loads highlights its proficiency in managing increasing computational tasks efficiently. The linear growth in CPU time suggests a robust system that scales effectively with rising demands. Compared to Computers A and B, Computer C demonstrates a slight edge in efficiency, particularly in handling mid-range loads.Its ability to maintain consistent increases in CPU time without unexpected spikes underscores its reliability and suitability for applications requiring scalable performance.In conclusion, Computer C proves to be highly capable of efficiently managing increasing computational loads. Its consistent performance and slight advantage in efficiency over Computers A and B at mid-range loads emphasize its robustness for diverse and demanding computational tasks.

To summarize, Computer C proves to be highly capable of efficiently managing increasing computational loads. Its consistent performance and a slight advantage in efficiency over Computers A and B at mid-range loads emphasize its robustness for diverse and demanding computational tasks. The results suggest that Computer C is the most suitable option for applications requiring scalable performance, while Computers A and B may need optimization to improve their efficiency.

4.0 Conclusion

Throughout this benchmarking project, we have conducted a comprehensive evaluation of the CPU performance of three different processors which are the Intel Core i5 1135G7 in Asus Vivobook, Intel Core i5 11300H in LENOVO 82K1 and Intel Core i5 11400H in Dell Inspiron 16. Our analysis involved using industry-standard tools such as CPU-Z, Geekbench, Window Task Manager and MiniTool Partition Wizard Free 12.8 as well as executing intricate assembly language loops to simulate real-world processing scenarios.

Based on the results from part 1 and part 2, it is clear that Intel Core i5 11400H in Dell Inspiron 16 is a powerful and efficient computing solution. It highlights the most impressive results from different aspects, such as single-core or multi-core performance, sequential or random writing, memory bandwidth and clock speed is particularly noteworthy, making it an ideal choice for several demanding tasks such as gaming, scientific simulations, video editing and data compression. Intel Core i5 11400H is a strong and versatile computing solution that can handle a wide range of tasks and applications. Its impressive performance and efficient resource utilization make it a valuable investment for users who require a high-performance computing solution.

In conclusion, this benchmarking study has not only provided a detailed comparison of the Dell Inspiron 16, Lenovo 82K1, and Asus Vivobook but has also underscored the significance of benchmarking in the pursuit of technological excellence. By providing insights on the strengths and weaknesses of each computer, benchmarking helps developers understand their product’s performance, allows them to refine their designs, helps end-users to make informed purchasing decisions, and the industry as a whole to drive innovation and progress. The importance of benchmarking will only continue to grow in this evolving computing landscape, serving as a beacon for improvement and a catalyst for the development of faster, more efficient, and more powerful technologies.

5.0 Reflection (each member)

Tan Zhao Hong

Due to unforeseen circumstances, I did not conduct benchmarking on my computer as my computer unexpectedly crashed and became unusable. Hence, I had to rely on my groupmate's results to write the report. By discussing the benchmarking methods and results, I was able to learn from their hands-on experience, which enhanced my understanding of the tools and processes involved. Besides, another challenge that I face is interpreting the results of the benchmarking test. I need to identify the most relevant metrics to compare the performance of the processors. This process made me realize that I need to develop my analytical skills so I can critically evaluate the technical data, then draw meaningful conclusions, and finally present them clearly in the report. Moreover, I also wish to express my gratitude to my lecturer, Dr. Fo'ad who provided guidelines for us to complete the project. This project has been a valuable learning experience that has enhanced my understanding of CPU performance and the importance of benchmarking in the pursuit of technological excellence. After this, I plan to conduct my benchmarks to gain firsthand experience when I get my computer. This will not only enhance my technical skills but also provide a more in-depth understanding of the process and results.

Chin Pei Wen

Through this project, I have gained a deeper understanding of the intricacies of CPU performance analysis and benchmarking. By examining cache configurations, memory specifications, motherboard features, and CPU performance across various loop counts, I gained insights into how these factors influence overall system efficiency and scalability. Additionally, analyzing execution times under varying workloads provided insights into the efficiency and scalability of different CPUs. This project has reinforced the value of meticulous data collection and comparison in identifying subtle differences in performance, which can inform decisions on hardware selection and optimization for specific tasks and workloads. Specially, I would like to express my sincere appreciation to my lecturer, Dr. Fo'ad, for providing clear instructions and offering unwavering support throughout the development of this project. His guidance has been invaluable and greatly contributed to our learning experience. Additionally, I want to extend my gratitude to my group members for their support and assistance. Their collaboration and dedication were instrumental in successfully completing this project. Thank you all for contributions and encouragement.

Koo Xuan

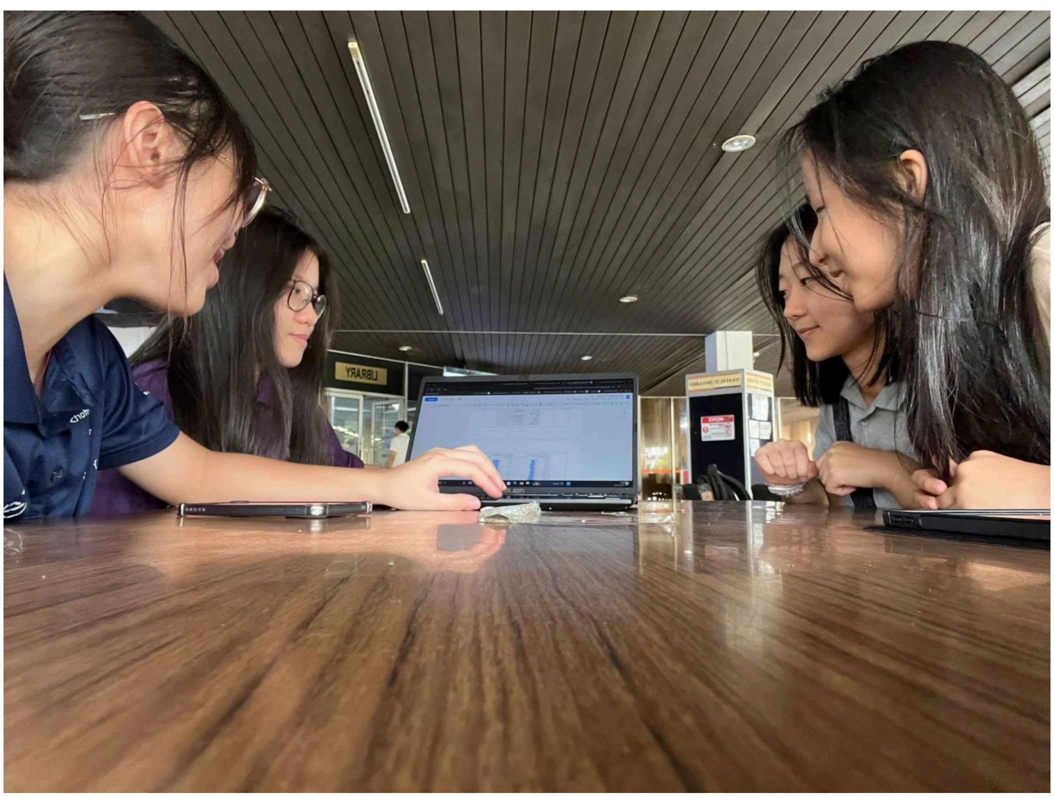
Throughout doing this project, I have learned how to use those tools such as CPU-Z, Geekbench, MiniTools and Task Manager to help me know more about aspects of computer systems and performance benchmarking. I gained a deeper understanding of my own computer system's capabilities and limitations, which was insightful and informative. Additionally, I learned how to effectively use Visual Studio for coding and testing, enhancing my technical skills and confidence in working with development tools. This project would not have been successful without the collaboration and support of my groupmates, whose contributions and teamwork were invaluable. I would also like to express my gratitude to our lecturer, Dr. Mohd Fo'ad Bin Rohani, for his guidance and expertise throughout his lecture. His support and encouragement were instrumental in helping us achieve our objectives and complete the project successfully.

Ling Yu Qian

In this project, I have developed valuable skills in using various tools like CPU-Z, Geekbench, MiniTools, and Task Manager, which provided me with a comprehensive understanding of computer systems and performance benchmarking. This hands-on experience allowed me to delve deeper into my own computer system's capabilities and limitations, offering insightful and informative revelations. Additionally, mastering Visual Studio for coding and testing has significantly enhanced my technical proficiency and confidence in utilizing development tools. This project's success is largely attributed to the collaborative efforts and unwavering support of my groupmates, whose contributions were indispensable. I am also profoundly grateful to our lecturer, Dr. Mohd Fo'ad Bin Rohani, for his exceptional guidance and expertise. His support and encouragement were pivotal in helping us achieve our objectives and complete the project successfully. This journey has been enriching and has equipped me with both practical skills and a deeper appreciation for teamwork and mentorship.

6.0 Appendix

A. Group Pictures



B. Task Division

Introduction (Chin Pei Wen)

Part 1 -Overview (Ling Yu Qian)
-Flowchart (Ling Yu Qian)
-Benchingmarking Result (Chin Pei Wen, Koo Xuan, Ling Yu Qian)
-Analysis, comparison, discussion : CPU-Z (Chin Pei Wen)
Geekbench (Tan Zhao Hong)
MiniTools (Koo Xuan)
Task Manager (Ling Yu Qian)

Part 2 - Overview (Tan Zhao Hong)
- Flowchart (Tan Zhao Hong)
- Coding and documentation (Koo Xuan)
- Implementation/execution and results (Chin Pei Wen, Koo Xuan, Ling Yu Qian)
- Analysis, comparison, discussion : Opening and closing (Tan Zhao Hong)
Computer A (Chin Pei Wen)
Computer B (Koo Xuan)
Computer C (Ling Yu Qian)

Conclusion (Tan Zhao Hong)

C. Proof for Crashed Computer

Note:

One of our group's computers crashed and became unusable. As a result, we conducted the benchmark using only three computers instead of four.